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Notes on Measure Theoretic Probability

The most important scientific tool of all is not anything you can buy. It is your own mind. Your thoughts and ideas are the keys that can unlock the mysteries. In the search for understanding, *questions* are perhaps the most powerful *force* of all.

— "The hidden world of forces" by Jack R White

PART A :THE THEORY

0. Getting Started

Before beginning

Is every continuous function differentiable?
Does every vector space have a basis?
Will it rain tomorrow?
Is it safe to release Nimuselide in the market?

You see, there are so many questions that one can ask and any intelligent person would be interested in the answers. To just guess and give a random answer without being able to explain why we felt so is one way of using our *instincts* and answer questions. A better way would be to understand the nature of things, develop a method of analyzing and then arrive at answers. With such a method we would not only be able to justify the answers but also if we are wrong, we can precisely pinpoint where we went wrong and modify our basic assumptions/methods. Such failures will help understand the nature of things we are modelling. This is the spirit of scientific inquiry and this is what scientists do. But each scientific discipline has its own methods.

For example the first two questions above have precise answers : 'no' for the first and 'yes' for the second. We are able to arrive at the precision because mathematicians made definitions of terms like 'continuous' 'basis' etc and the questions refer to those concepts. He made certain rules of inference and if the statement follows from those rules he says the answer is yes. If he can produce an example where the hypothesis is correct but the conclusion is not correct he says the statement is not true. Of course this does not mean that things are straightforward or routine – far from it. Still there are several questions that defy analysis.

The third questions is not so simple. You can make good definitions for 'temperature' 'wind speed', 'cloud concentration', 'atmospheric pressure' etc, but you can not make any precise rule as to what values of these parameters will surely lead to rain. Secondly, the question refers to tomorrow, not today. This makes it more difficult to answer. Nonetheless, you must agree that this question is just as important as the first two questions. If there is going to be heavy rain tomorrow and if you can predict it today, the farmers/fishermen will benefit.

The last question is very important for the society. If the medicine is useful and you do not market it you are depriving the society of its benefits. On the otherhand, if it is harmful and you market it you are causing serious damage to the society. But how do you answer the question? (Incidentally, for this last question, no one is sure what is right answer, though this medicine is used often).

For questions of this later kind, you have to build some models and see if they are good models that fit the past observations of the phenomenan (or the observed experimental data). Then try to develop a method of prediction and finally predict. This is what a statistician does. His job is much more difficult than the others. Mainly because he is dealing with certain phenomena that depend on what can be called chance. Naturally, there is every chance of his predictions going wrong. But if they are correct in most cases we can be happy that we found a satisfactory method of answering our questions.

In probability theory we develop methods of building models of chance phenomena. Hopefully, they will be useful and one of the models fits the data at hand and helps us. So you, as students, should develop a certain understanding as to what kind of questions we are trying to tackle and how we are doing it. Unfortunately, it has become fashionable to come to instantaneous judgements 'this is rigorous', 'this is not rigorous', 'this is simple', ' this is abstract' etc without even understanding what we are aiming at.

Generally scientists seem to keep track of the foundations of their areas. For example, a physicist, no matter in which topic he works, shows interest and keeps track of the basic foundations of his theory. We mathematicians/statisticians seem to have lost this cultural aspect and seem to be confined to certain narrow boundary walls. I am not sure how this happened. I do not think your generation should continue this way. Measure theory is neither difficult nor abstract. What do we need, in order to learn this? Just one thing, as Paul Erdos was fond of saying, 'keep your brains open'.

The beginning

The basic aim of Probability Theory is to analyze experiments that depend on chance. The first reaction is that if the experiment depends on chance, can we say anything at all? Yes, there is some regularity/beauty even in chance phenomena! What do we mean by analysis? We ask simple questions and answer them. For example, if I shuffle a pack of cards and deal 13 of them to you, how likely is it to get all four aces? Did you say it depends on luck? You are right. But if we leave it at that then we are not taking up the challenge to answer simple questions. Did you say this is a useless activity to shuffle cards and deal them. I do not agree. But even if you think so, let me add that this is only by way of illustration. Analysis of such experiments applies not only to a pack of cards but essentially to any similar experiment. There are many other things that we talk about later. Did you say that the chances are very small? Yes, but if you leave it at that it is no good. We are supposed to be intellegent. We should try to give a quantitative answer.

So what do we need in order to undertake analysis of chance experiments? The first thing is to understand the experiment, in particular, have a clear idea of the possible outcomes of the experiment. The set of such possible outcomes is denoted by Ω whose elements are to be denoted by ω. The set Ω is also called

the *sample space* of the experiment and its elements ω are called *sample points* or *outcomes*. Of course, there are several interesting possibilities. If you think of the example of card deal mentioned above, there are finitely many outcomes (though the number is huge). Suppose you toss a coin until you get heads and then stop. Eventhough you are sure to stop at a finite stage (unless both sides of the coin are tails, which we do not allow), there is no *a priori* bound on the number of tosses needed; you can see that there are infinitely many possible outcomes. Yet the number of outcomes is countable. There are different sort of experiments that are necessary for real life modelling situations. You want to model the lifetime of a bulb, measured in some prespecified units. You agree that any non-negative number can be a possible value. So, the sample space should consist of all non-negative numbers.

In the earlier courses on probability you dealt with discrete experiments, that is, experiments where the set Ω is a finite or countably infinite set. For every outcome ω you associated a number $p(\omega)$ called the chance of the outcome. For obvious reasons these numbers are non-negative and add upto 1. An *event* was any set of out comes, that is, any subset $A \subset \Omega$ was an event. For an event A, we associated its probability $P(A)$ by just defining it as sum of all the probabilities of out comes in this event; in symbols, $P(A) = \sum_{\omega \in A} p(\omega)$. Things went on smoothly.

Let us consider the experiment of choosing a number at random from the interval $(0, 1]$. Here, every number in this interval is a possible outcome so that $\Omega = (0, 1]$, a set which is uncountable. If we follow the same simplistic philosophy of assigning numbers $p(\omega)$ to each outcome, we run into difficulties as follows. Fix any number x in $(0, 1]$. Since we are selecting a point at random the chances of it falling in $(0, 1/2]$ and the chances of it falling in $(1/2, 1]$ must be the same and hence each must be equal to $1/2$. Whether x is in $(0, 1/2]$ or $(1/2, 1]$ you conclude that $p(x) \leq 1/2$. You can continue this arguemnt by splitting this interval again to see $p(x) \leq 1/4$ and so on, yielding $p(x) = 0$. Thus if you try to associate probabilities to outcomes you observe that each outcome must have probability zero.

We can take clue from the concept of length. You see that points have length zero, but intervals have non-zero length. Perhaps we should not try to associate probabilities to outcomes, we should try to associate at one stroke a probability to each event. But then what are events? Imitation being the first choice (also due to lack of any guiding principle at this stage), let us say that every subset of $\Omega = (0, 1]$ is an event. Thus for every subset A of the unit interval we want to associate a number $P(A)$ which represents the probability of the event – that is $P(A)$ is the chance that the selected number belongs to the set A.

There are some natural demands to be satisfied. For example $P(\emptyset) = 0$ and $P(\Omega) = 1$. Further if we have disjoint events $A_n, n \geq 1$ with $\cup A_n = A$ then we should have $P(A) = \sum P(A_n)$. This condition is called countable additivity condition. These conditions are dictated by our intuitive notion of what a probability ought to be. Now there is another condition that reflects the fact that we want to model the specific experiment of picking a point at random. As noted earlier this amounts to demanding that for any dyadic interval $(k/2^n, (k+1)/2^n]$ the value of the probability should be $1/2^n$. If we can make an assignment of a value $P(A)$ for each event A so that these conditions are satisfied we will have a model for the experiment and can feel happy. Unfortunately our

task is a near impossibility due to a theorem proved by the Polish Mathematician S.M.Ulam.

Theorem 1: Impossibility of model

(Assume two set theoretical hypotheses : Axiom of choice and Continuum Hypothesis) There is no function P defined for all subsets A of the unit interval such that it is non negative, countably additive, and for dyadic intervals $P(k/2^n, (k+1)/2^n] = 1/2^n$.

Do not bother about the set theoretical hypotheses mentioned. This is not too important for our present discussion. The moral is that our attempts fail. At this juncture you will surely be tempted to ask : please name one set A for which we fail to associate probability. No, I can not do that. The condition of countable additivity is a collective condition, not an individual condition applying to each event A. This causes problems. To illustrate with a simple analogy, you are 16 students in the class. Suppose that I have 31 apples. I say I can not give two apples to each of you. You will surely agree. But can I name one single person to whom I fail to give? No, simply because any person you name I can surely start giving two apples to him.

Where did we go wrong? Which of our demands is so tough that it can not be met? Perhaps we should not demand countable additivity. But then we will run into deep troubles. We will have problems with what we are used to so far. Remember how we calculated the probability that a Poisson random variable takes an even integer value. We added up the probabilities of even integers. Inspite of this, it is perfectly legitimate for you to say that we abandon countable additivity. You may well argue that the Poisson example mentioned above are special cases - - afterall, demanding finite additivity does not exclude the possibility of countable additivity. You are very right. There are strong arguments in favour of abandoning countable additivity and instead demand only finite additivity. Attempts at developing models with only finite additivity were performed. The upshot seems to be that such a theory is messy, mathematically difficult, a not so fertile ground for analysis and shows cracks when trying to model independent repetitions of random experiments.

The only demand on which we have to compromise is 'every subset is an event'. Is it really necessary? What exactly is the purpose of our building a mathematical model? We want to answer simple questions concerning the experiment. So if we can assign probabilities to all the subsets that we are likely to come across in practice then it should serve the purpose. Thus we abandon the assumption that every subset be an event.

Then the question is : which subsets should be events? Clearly each interval should be an event. If A is an event then its complement A^c should also be an event. (If you can answer 'what are the chances of something happening', then we should be able to answer 'what are the chances of that something not happening'). If we have countably many events A_n then their union $A = \cup A_n$ should also be an event. (if some one asks you 'what are the chances of this happening' 'what are the chances of that happening' etc then be prepared that he may ask you tomorrow 'what are the chances of one of the things that I mentioned yesterday happening'). Let $\mathcal{B}$ be the smallest class of subsets of Ω satisfying these conditions. We agree to treat sets that belong to this class as events. Sets which are not in this class are not events.

With this change of attitude, to build a mathematical model we need to assign probability $P(A)$ to all events A – that is, to sets belonging to the class $\mathcal{B}$. This assignment should satisfy the same demands as earlier.

Lo and Behold, we can build a model!

Theorem 2 : Possibility of a model

There is a function P defined for all sets in $\mathcal{B}$ such that it is non negative, countably additive, and for dyadic intervals $P(k/2^n, (k+1)/2^n] = 1/2^n$.

Success at last, but the price we have to pay is that not every subset of sample space is regarded as an event. This is not as bad as it looks like. How many subsets of $(0, 1]$ are there? Toooooo many. How many sets are we likely to come across in practice? Verrrrry few.

One doubt may still linger in your mind. Why not associate probabilities only for intervals and their countable unions and stop. Go back and read the reasons for our demand that the collections of events be closed under complements and countable unions. These operations on events are necessary to answer reasonable questions that an experimenter may ask. For instance, in the example of picking a point at random from $(0, 1]$, the experimenter may ask: what are the chances that in the decimal expansion of the number picked all digits 0 to 9 appear with equal frequency. This is a simple question which should be given a meaning and answered. Unfortunately, countable union of intervals will not describe such events.

The question that now arises is : who decides what subsets should be called events? But if you think about it, it should be clear to you that it depends on the experiment under consideration and the questions we are likely to be interested. So there is no decision that can be made *a priori*. We can develop general theory and depending on specific experiment we can specialize the theory.

1. Probability Spaces

The basic set up

This brings us to the basic setup for measure theoretic probability. We should have a triple $(\Omega, \mathcal{A}, P)$ where Ω is a non-empty set – describing the possible outcomes of the experiment under consideration; $\mathcal{A}$ is a family of subsets of Ω closed under countable unions and complements and containing Ω and empty set – sets in this class are to be called events; and finally a function P defined on $\mathcal{A}$ which is nonnegative and countably additive with $P(\Omega) = 1$ – describing the probabilities of events. This setup is general enough to allow discussion of all chance experiments. Such a triple is called a *probability space*. Such a family of subsets $\mathcal{A}$ is called a *σ-field* of subsets of Ω.

The first order of business is to see examples of σ-fields and to explain how to construct σ-fields.

For any Ω the family of all subsets is a σ-field of subsets of Ω. At the other extreme is the family consisting of just the two sets: $\emptyset$ and Ω. This family is also a σ-field. You can think of several in between. For example take any set $A \subset \Omega$ and consider the family that consists of the four sets: $\emptyset$, A, A^c and Ω. More generally take any finite or countable partition of Ω and consider the

family which consists of precisely those sets that can be obtained as union of some sets in the partition.

Here is another one. Consider those sets which are countable *or* those sets whose complements are countable. This family is also a σ-field. This is called the countable-cocountable σ-field. Cocountable set means complement of a countable set and hence this nomenclature. Of course if Ω itself is countable , then this family indeed consists of all subsets of Ω. However if Ω is uncountable then this does not consist of all subsets.

Here is more satisfying answer to constructing σ-fields.

Theorem 3A: Getting σ-fields

Let Ω be any non-empty set and let $\mathcal{C}$ be any family of subsets of Ω. Then there is a σ-field $\mathcal{A}$ of subsets of Ω which includes all sets from the given family $\mathcal{C}$ and which is smallest such. That is, if $\mathcal{B}$ is any σ-field of subsets of Ω that includes the given family $\mathcal{C}$ then $\mathcal{A} \subset \mathcal{B}$.

Clearly the σ-field $\mathcal{A}$ of the theorem above is unique. It is called the σ-field generated by the family $\mathcal{C}$ denoted by $\sigma(\mathcal{C})$. You will see theorem 3B after theorem 8.

For example you can take R for Ω and the collection of all intervals for the family $\mathcal{C}$. The resulting σ-field of subsets of R is called the Borel σ-field on R and sets in this family are called Borel sets. No matter how hard you try, you can not think of a set which is not a Borel set. Of course this does not mean that all subset of R are Borel – far from it. There are sets which are not Borel. How does one see this? There are two natural ways. First is to count how many Borel sets are there and show that this number is strictly smaller than the number of all possible subsets of R. This needs the concept of *cardinal numbers*. Second method is to explicitly construct a set which is not Borel using the well-known Cantor diagonal argument. This needs considerable work. We attempt neither at this stage. To bring in a simple analogy, if you have counted that you have 200 fruits and if you also counted that the number of apples among them is 30, then surely there are fruits which are not apples. Of course, in dealing with finite sets it is perhaps easier to pick up one fruit which is not apple and conclude that there are fruits which are not apples in our collection.

Eventhough the above theorem is satisfying, we should note that in general there are many many sets in the σ-field generated by a given family $\mathcal{C}$ – just like the collection of Borel sets of R. It is not easy to *see* all of them. The natural question that arises is : If I do not see all the sets in my σ-field how am I going to handle them and in particular define probabilities? Coming to think of it, aren't there too many real numbers – too many to be listed. Nonetheless we had no problem in doing Calculus. There are several ways of handling σ-fields. Here is one way.

Say that a family $\mathcal{F}$ of subsets of Ω is a *field* if the following properties hold: (i) $\Omega \in \mathcal{F}$ and $\emptyset \in \mathcal{F}$; (ii) if $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$; and (iii) if A and B are in $\mathcal{F}$ then their union $A \cup B$ is also in $\mathcal{F}$.

Thus a field differs from a σ-field only in one aspect – namely, a field is closed under finite unions whereas a σ-field is closed under countable unions. This actually makes a lot of difference. Given any family $\mathcal{C}$ of subsets of Ω there is a smallest field $\mathcal{F}(\mathcal{C})$ that includes the given class. It is interesting to note that we can give an algorithm (in exercises) which in a finite number of steps

constructs $\mathcal{F}(\mathcal{C})$ from $\mathcal{C}$.

Say that a family $\mathcal{M}$ of subsets of Ω is a *monotone class* if the following holds: if $A_n \uparrow A$ or $A_n \downarrow A$ and each $A_n \in \mathcal{M}$ then $A \in \mathcal{M}$.

Here $A_n \uparrow A$ means that $A_1 \subset A_2 \subset A_3 \subset \cdots$ and $A = \cup_n A_n$. And $A_n \downarrow A$ means that $A_1 \supset A_2 \supset A_3 \supset \cdots$ and $A = \cap_n A_n$.

For example if Ω is the set of real numbers then the collection of all intervals (empty interval, degenerate intervals, open, closed, semiopen) is a monotone class. Again given any class $\mathcal{C}$ of subsets of Ω there is a smallest monotone class $\mathcal{M}(\mathcal{C})$ of subsets of Ω that includes the given class.

Theorem 4 : Monotone Class Theorem, MCT
If $\mathcal{F}$ is a field of subsets of Ω then $\sigma(\mathcal{F}) = \mathcal{M}(\mathcal{F})$.

How does this help us? First let us observe a property of probabilities.

Theorem 5 : Continuity of Probability
Let $(\Omega, \mathcal{A}, P)$ be a probability space. Let (A_n) be a sequence of events (that is, sets in $\mathcal{A}$). If $A_n \uparrow A$ then $P(A_n) \uparrow P(A)$. If $A_n \downarrow A$ then $P(A_n) \downarrow P(A)$.

Conversely, suppose P is a finitely additive probability on $\mathcal{A}$, that is, P is defined on $\mathcal{A}$ with values in $[0, 1]$, $P(\emptyset) = 0, P(\Omega) = 1$ and for disjoint sets A and B in $\mathcal{A}$, $P(A \cup B) = P(A) + P(B)$. Suppose that $P(A_n) \downarrow 0$ whenever $A_n \downarrow \emptyset$. Then P is countably additive and is hence a probability.

Theorem 6: A Uniqueness Theorem for Probabilities
Let $\mathcal{F}$ be a field of subsets of Ω and $\mathcal{A} = \sigma(\mathcal{F})$. Suppose that P and Q be two probabilities on $\mathcal{A}$. If they agree on $\mathcal{F}$ then they agree on $\mathcal{A}$.

In otherwords if $P(F) = Q(F)$ for every set $F \in \mathcal{F}$, then $P(A) = Q(A)$ for every set $A \in \mathcal{A}$. The beauty is that you need not have a list of all sets in the σ-field before you, nor is it necessary for you to see all sets in the σ-field. Usually fields are simple objects and explicit calculations can be made. That is, you can actually catch hold of all sets in $\mathcal{F}$ and show that the hypothesis is true. If you did that then there is no need for you to catch hold of all sets in the σ-field to arrive at the conclusion. We will see more applications of the MCT as we go along.

How to construct probabilities

It is easy to make definitions. But how do we construct probabilities. Even if we take Ω to be the real line and $\mathcal{A}$ to be its Borel σ-field, how are we ever going to construct probabilities on this σ-field. Remember to construct probability we have to assign a number $P(A)$ to each Borel set A. We must also show that it is countably additive. Definitely a tall order – especially under our present circumstances where we did not even see all the Borel sets. You are sitting in this room with fans. Is it necessary to see the electrons to make them work for you. You only have to cleverly devise methods that can be implemented. This is luckily possible with our definitions. First let us make one more useful definition.

Suppose that $\mathcal{F}$ is a field of subsets of Ω and P be defined on $\mathcal{F}$. Say that P is a probability on $\mathcal{F}$ if the following hold: (i) $P(\emptyset) = 0$ and $P(\Omega) = 1$; (ii) Whenever we have a sequence A_n of pairwise disjoint sets in $\mathcal{F}$ whose union A

is also in $\mathcal{F}$, then $P(A) = \sum P(A_n)$. Note that as opposed to the definition of countably additivity on a σ-field, here, in case of a field, we have to say that if each A_n is in the field *and also that their union is in the field* then we demanded countable additivity, of course, if the sets are disjoint. This is because it is quite possible that the union may not be in the field.

Theorem 7: Caratheodory Extension Theorem
Let $\mathcal{F}$ be a field of subsets of Ω and $\mathcal{A} = \sigma(\mathcal{F})$. Suppose that P is a probability on $\mathcal{F}$. Then P has a unique extension as a probability to $\mathcal{A}$.

This means that we have a probability P_1 on $\mathcal{A}$ such that for sets $F \in \mathcal{F}$, $P_1(F) = P(F)$. Further there is only one such extension. Usually the extension is also denoted by P. Yes, mathematically speaking the domain of definition of P is sets in $\mathcal{F}$ whereas the domain of definition of P_1 is sets in $\mathcal{A}$. So there is a clear distinction. But you agree that notation should help us to understand things rather than being a burden. As long as you and I understand what we mean (and can successfully communicate with others), it is fine.

What is the importance of this theorem for us? It tremendously reduces our job of constructing probabilities. If you want to construct a probability on a σ-field $\mathcal{A}$ then you need not really catch hold of each set in $\mathcal{A}$ and prescribe $P(A)$. You choose *any* convenient field $\mathcal{F}$ so that $\sigma(\mathcal{F}) = \mathcal{A}$ and just construct probability on the field you have chosen. The theorem above assures that there is then a unique probability on $\mathcal{A}$ which agrees with what you prescribed on the field. Thus you did indeed exhibit a probability on the σ-field itself. Thanks to Caratheodory, our job of constructing probabilities on a σ-field is very much simplified.

The question still remains. Can you execute this idea in any concrete case? Yes. This is what we do now. What is more concrete than R and its Borel σ-field?. We shall now go to our earlier probability course and recall the definition of distribution function.

A function F defined on R is said to be a *distribution function* (*d.f.* for short) if F is monotone nondecreasing, right continuous with limits 0 at $-\infty$ and 1 at $+\infty$.

Suppose that P is a probability on the Borel σ-field $\mathcal{B}$ of R. Put $F(x) = P(-\infty, x]$. Then F is a *d.f.* The beauty is that the converse is also true.

Theorem 8: Basic Existence Theorem of Probability Theory
Given a *d.f.* F on R, there is a unique probability P on $(R, \mathcal{B})$ such that for every $x \in R$, $F(x) = P(-\infty, x]$.

This is a very satisfying theorem for several reasons. Firstly, it produces examples of probabilites on the Borel σ-field of the real line. You only need to take a *d.f.* and you have a probability associated with it on the Borel σ-field. Secondly, it produces *all* probabilities on the real line. Caratheodory already reduced our job of constructing probabilities on a σ-field. He assured us that if we can do it on a field that generates the σ-field then he will help us extending it to the entire σ-field. This theorem reduces the job still further. Do not bother about field. Just give me a *d.f.*, that is enough. Then we can show a probability on the Borel σ-field. It is easy to construct *d.f.*s on R. In fact you have come across several of them in earlier course, like the normal, uniform, exponential beta, gamma, Binomial, Poisson, etc etc. You can cook some new distribution

functions like the Canor distribution function (in exercises).

2. Random Variables

Thus the ingredients for chance experiment consist of a set Ω which describes the set of al possible outcomes of the experiment; a σ-field $\mathcal{A}$ of subsets of Ω which consists of the collection of events, and a (countably additive) probability P defined on the collection of events. The next question is : what should be a random variable. Generally, we perform an experiment and make a measurement. This measurement is called random variable. For example the experiment may consist of tossing a coin 20 times, and you may want to count the number of heads obtained. Here the number of heads is the random variable. Thus random variables are nothing but measurements – mathematically, real valued function defined on the sample space Ω.

In dealing with general chance experiments, we have to be a little more careful. We may endup with objects about which we can not answer even simple questions. For example, suppose X is a measurement – that is, a real valued function on Ω. We are interested in the chances that the measurement is no more than 23. To calculate this, we should collect the set A of all sample points for which the measurement is at most 23 and then calculate the probability of this event A (Remember how we calculated the chances that there are exactly 5 heads in 20 tosses of a fiar coin). This is fine, but the only problem is that I have already used the word event for the set A. How do I know that this set A is in my σ-field? There is nothing so far to tell us that $A \in \mathcal{A}$. In fact it need not be. In that case we can not answer the simple question raised above. Is there any point admitting such measurements, if we can not answer even simple questions regarding them. We shall not admit such measurements as random variables. Thus for us a *random variable* is a real valued function X defined on Ω such that for every real number a, the set $\{\omega : X(\omega) \leq a\}$ is in $\mathcal{A}$.

There are several questions/doubts that arise immediately. First, how do you know that your demand above is good enough to answer all possible simple questions. An easy argument shows that when the above condition holds, then in fact, for every Borel set $B \subset R$, we have $\{\omega : X(\omega) \in B\} \in \mathcal{A}$.

Second, my measurement X is very important to me, does not satisfy your requirement, how can you exclude it from discussion? The point is well made. But then the answer is : your choice of σ-field is wrong. Your initial choice of events does not take into account the fact that you need to answer questions regarding this particular X. You should start with a better σ-field which makes X a random variable.

Third, how do I know that you are admitting enough random variables or not? Imagine for a moment that our definition is so bad that only constant functions are random variables, no matter what $\mathcal{A}$ is. Well, there are enough random variables if there are enough events.

Finally, how do I know that the definition has reasonable properties. For example if you have admitted two measurements for discussion (as random variables) it is only fair that their sum, which is again a measurement based on my experiment, should be admitted too.

Here are two theorem that answer all these and other questions in a satisfactory way.

Theorem 3B : Getting σ-fields

Let Ω be any set and Φ be any collection of real valued functions on Ω. Then there is a σ-field $\mathcal{A}$ of subsets of Ω such that each function in Φ is a random variable w.r.t this σ-field and it is the smallest such.

Clearly, the σ-field of the above theorem is unique. It is named as the σ-field generated by the class Φ, denoted $\sigma(\Phi)$.

Suppose now that $(\Omega, \mathcal{A}, P)$ is a probability space and L is the collection of all real random variables. By properties of inverse images, clearly those sets $B \subset R$ such that $X^{-1}(B) \in \mathcal{A}$ is a σ-field and if it includes sets of the form $(-\infty, a)$ then it includes all Borel sets. As a result, $X \in L$ iff for every Borel set $B \subset R$, we have $X^{-1}(B) \in \mathcal{A}$. Of course, the concept depends only on the σ-field and not on the probability P (as it should be – because measurement depends on the sample points and not on what probability model is assumed). Among random variables let us identify the simplest ones : A random variable X is *simple random variable* if it takes finitely many real values. We denote by I_A the indicator function of the set A, namely the function which is 1 for points of A and zero for points not in A.

Theorem 9 : Structure of Random Variables

(i) The space L is a real vector space, that is, if X and Y are in L, then so is $\alpha X + \beta Y$ for any real numbers α and β. L is an algebra, that is, besides being a vector space it is closed under multiplication. L is a lattice too, that is, the (pointwise) maximum and minimum of two random variables is again a random variable. Constant functions are in L. Thus in particular, for any random variable X, both $X^+ = Max(X, 0)$ and $X^- = Max(-X, 0)$ are random variables.

(ii) If $X_n \in L$ and $X_n \uparrow X$ or $X_n \downarrow X$ pointwise and X is real valued, then $X \in L$. In particular pointwise lim sup and lim inf of X_n are in L provided they are real valued.

(iii) $I_A \in L$ iff $A \in \mathcal{A}$. The collection of simple random variables is also a vector space, algebra and lattice – as in (i) above.

(iv) Given any bounded random variable X there is a sequence (s_n) of simple random variables such that $s_n \uparrow X$ uniformly. Given any non-negative random variable X, there is a sequence (s_n) of simple non-negative random variables such that $s_n \uparrow X$ pointwise.

(v) Suppose that $\mathcal{F}$ is a field of sets such that $\mathcal{A} = \sigma(\mathcal{F})$. Suppose that Φ is a vector space of real valued functions on Ω closed under monotone limits — which means that if each $X_n \in \Phi$ and $X_n \uparrow X$ or $X_n \downarrow X$ pointwise and X is real valued, then $X \in \Phi$. Suppose that for each $A \in \mathcal{F}$, we have $I_A \in \mathcal{F}$. Then $L \subset \Phi$.

(vi) Suppose that $\mathcal{F}$ is a field of sets such that $\mathcal{A} = \sigma(\mathcal{F})$. Suppose that Φ is a vector space of real valued functions on Ω closed under bounded pointwise convergence — which means that if each $X_n \in \Phi$ and $X_n \to X$ or pointwise and X_n are uniformly bounded, then $X \in \Phi$. Suppose that for each $A \in \mathcal{F}$, we have $I_A \in \mathcal{F}$. Then every bounded random variable is in Φ.

It is rather awkward that we assumed in (ii) above that the lim sup etc are real valued. Actually this is unnecessary. But we have to admit extended real valued functions to make things precise. We are not doing this at present. This

will be taken care of later. The last two parts of the theorem above help us in dealing with the collection of all random variables just as the MCT helps us in dealing with all sets in the σ-field.

3. Expectation

What is expectation

Suppose that $(\Omega, \mathcal{A}, P)$ is a probability space. Suppose that s is a simple non-negative random variable say $s = \sum a_i I_{A_i}$ where for each i, $a_i \geq 0$. We define $E(s) = \sum a_i P(A_i)$. If X is a non-negative random variable then we put $E(X) = \sup\{E(s) : s \text{ simple }, 0 \leq s \leq X\}$. If X is any random variable, we say that X is integrable iff $E(X^+) < \infty$ and $E(X^-) < \infty$ and in such a case we put $E(X) = E(X^+) - E(X^-)$. We shall also write $\int X dP$ instead of $E(X)$. We denote the collection of integrable random variables by $L^1(\Omega, \mathcal{A}, P)$, in short, L^1 when other things are understood.

Why is it defined this way

The first question that should be answered is the reasons to adapt this definition. Suppose we have $s = I_A$, then our definition gives $E(s) = P(A)$. This stands to reason because s takes two values one and zero with probabilities $P(A)$ and $1 - P(A)$ so that, using what we learnt in earlier course, we should indeed have $E(s) = P(A)$. Since expectation is, in a sense, average value it should be linear. So for non-negative simple functions it should be as defined above. Another way of thinking of it is as follows: say s takes the distinct values $a_i \geq 0$ and let $A_i = \{\omega : s(\omega) = a_i\}$ so that $s = \sum a_i I_{A_i}$. Then as you can see the random variable s takes the value a_i with probability $P(A_i)$ and hence its expectation should be $\sum a_i P(A_i)$. This is what we get from our definition too. Of course what is not clear is whether any representation of simple function will lead to the same answer. That is, one needs to show that the formula for $E(s)$ does not depend on the representation used in the definition.

It is clear that if we have two measurements $X \leq Y$ then the average value of X should not exceed the average value of Y. Thus in particular for every simple non-negative $s \leq X$ we must have $E(s) \leq E(X)$. Thus $E(X)$ should not fall below the value we have prescribed. The best way at this stage is to see if we can get an upper bound for the possible value of $E(X)$ and then decide what to do. If luckily these two values coincide then there is nothing for us to decide. We did not do this (though this path can also be pursued carefully). Since there is no *a priori* reason why $E(X)$ should exceed this lower bound (if so by how much? etc) we have declared this lower bound as $E(X)$. If you take a general random variable X, note that $X = X^+ - X^-$ so that linearity justifies the definition we gave. Of course it is not clear whether the expectation so defined has reasonable properties at all. The next theorem says that all properties you expect from *expectation* hold with this definition.

Theorem 10: Properties of Expectation

(i) If a simple nonnegative random variable s is expressed in two different ways as $\sum a_i I_{A_i}$ with $a_i \geq 0$ as well as $\sum b_j I_{B_j}$ with $b_j \geq 0$ then we have $\sum a_i P(A_i) = \sum b_j P(B_j)$. In other words $E(s)$ for simple nonnegative random

variables is well defined. $E(s)$ is nonnegative linear in the sense that for simple nonnegative s and t and non-negative reals α and β we have $E(\alpha s + \beta t) = \alpha E(s) + \beta E(t)$. $E(s)$ is monotone in the sense that if $s \leq t$ then $E(s) \leq E(t)$.

(ii) The clause defining integral for non-negative random variables extends the definition given for simple non-negative random variables earlier. $E(X)$ is monotone on the collection of non-negative random variables.

(iii) (Monotone convergence theorem ; **MCT**) If $X_n \geq 0$ and $X_n \uparrow X$ then $E(X_n) \uparrow E(X)$.

(iv) If $X \geq 0$, $Y \geq 0$ and $\alpha, \beta \geq 0$ then $E(\alpha X + \beta Y) = \alpha E(X) + \beta E(Y)$.

(v) The final clause defining integral for general (integrable) random variables extends the definition for non-negative integrable random variables. L^1 is a vector space and expectation is linear on this space and is also monotone. Further, $X \in L^1$ iff $|X| \in L^1$. Finally, $|\int X dP| \leq \int |X| dP$.

(vi) (**Fatou's** lemma) $\int LimInf X_n dP \leq LimInf \int X_n dP$ for nonnegative random variables X_n,

(vii) (Lebesgue's Dominated convergence theorem ;**DCT**) If $X_n \in L^1$ for each n; $X_n \to X$ pointwise ; there is $Y \in L^1$ such that $|X_n| \leq Y$ pointwise for all n, then $X \in L^1$ and $\int |X_n - X| dP \to 0$ and $\int X_n dP \to \int X dP$.

(viii) For any non-negative random variable X, $P(X \neq 0) = 0$ iff for any set $A \in \mathcal{A}$, $E(XI_A) = 0$. As a consequence, for two random variables X and Y, $P(X \neq Y) = 0$ iff for any set $A \in \mathcal{A}$, $E(XI_A) = E(YI_A)$.

4. Some fine tuning

There are some important details that need to be attended to. There are no new ideas – only extensions of the ideas discussed so far.

infinities

We need to discuss random variables that may take values $\pm\infty$ also. Not that we invite them, but they may crop up in our calculations. We realized it while dealing with lim sup and lim inf of random variables. There are other reasons too. Suppose you take i.i.d normal observations and wish to study the properties of the observed averages. There is no reason why the observed averages can not go to ∞. In fact they do (of course with probability zero).

Extended real line means the set $\{-\infty\} \cup R \cup \{+\infty\}$, with the understanding that $-\infty < x < +\infty$ for all real numbers x. Usually $+\infty$ is also denoted as ∞. A function taking values in this set is said to be an extended real valued function. Such a function X is said to be an extended real random variable if for every real number x, $\{\omega : X(\omega) \leq x\} \in \mathcal{A}$ – as in the case of real random variables. We have the common sense arithmetic on this extended real line. for every real x, $\pm\infty + x = \pm\infty$; $\infty + \infty = \infty$; $-\infty - \infty = -\infty$. We agree NOT to talk about $\infty - \infty$.

With this arithmetic we can define sums of two extended random variables provided at no sample point one of them assumes value ∞ and the other assumes the value $-\infty$. Then the sum will also be an extended real random variable. Limsups and lim infs are all (extended real) random variables. Integration can also be defined. Of course one still takes the same old definition for simple random variables – that is, the values $\pm\infty$ are not allowed for simple random variables.

σ-finite measures

Sometimes it is necessary to look at things defined on σ-fields which do satisfy countable additivity and are non-negative but the value for whole space is not necessarily one.

Suppose that Ω is a set and $\mathcal{A}$ is a σ-field of subsets of Ω. A function μ defined on $\mathcal{A}$ with values in $[0, \infty)$ is said to be a finite measure if $\mu(\emptyset) = 0$, and μ is countably additive. Clearly if $\mu(\Omega) = c$ then for every set $A \in \mathcal{A}$ we have $\mu(A) \leq c$.

A function μ defined on $\mathcal{A}$ with values in $[0, \infty]$ is said to be a σ-finite measure if $\mu(\emptyset) = 0$, μ is countably additive and Ω is a union of countably many sets in $\mathcal{A}$ each having finite measure – that is, each having finite μ value.

The extension theorem of Caratheodory is immediate for finite measures. It is also easy to deduce it for measures which are σ-finite on the field $\mathcal{F}$ – that is, when μ given on $\mathcal{F}$ is such that there exists a sequence of sets in $\mathcal{F}$ of finite μ value whose union is Ω then it can be extended to $\sigma(\mathcal{F})$. Since the definition of random variable does not depend on the probability/measure there is no new problem. We can define integral as in the case of probability and all those theorems are valid. But of course some care is needed. For example if μ is σ-finite then $A_n \downarrow A$ need not imply $\mu(A_n) \downarrow \mu(A)$. Similarly if μ is not finite (but σ-finite) then constant function $\mathbf{1}$ is not integrable whereas on a probability space the constant functions are always integrable.

There is a unique σ-finite measure λ on the Borel σ-field of the real line such that for intervals it coincides with length – that is for $a < b$, we have $\lambda(a, b] = b - a$. This is called the Lebesgue measure. For instance, the standard normal density is a density w.r.t this measure.

complex random variables

If you have a random variable then you learnt that $M(t) = E(e^{tX})$ is the moment generating function of the random variable X. The only problem is that the required expectation may not exist and hence this function may not be defined for all values of t. However if we change slightly, $\varphi(t) = E(e^{itX})$ always exists and as you know this is called the characterstic function of the random variable$X$. This function helps in the study of the random variable X and in particular in the Central limit theorem. In fact this will put at our disposal all the results of complex function theory. So there is advantage in discussing complex valued functions.

Note that if X is a complex valued function on a set Ω, then there are two uniquely defined real valued functions X_1 and X_2 such that $X(\omega) = X_1(\omega) + iX_2(\omega)$. If we have a σ-field $\mathcal{A}$ on Ω then we say that X is a random variable iff both X_1 and X_2 are so. We already know when to say a real valued function is a random variable. If moreover we have a probability P on $\mathcal{A}$ then we say that X is integrable if both X_1 and X_2 are so and then we define $E(X) = E(X_1) + iE(X_2)$.

All the results for expectation remain valid. Of course there is no linear order on complex numbers and hence MCT does not make sense. But the fact $|E(X)| \leq E|X|$ and the DCT remain valid.

almost everywhere

While dealing with random variables we should not ignore the existence of probability. For instance $X \leq Y$ if for every sample point ω, $X(\omega) \leq Y(\omega)$ is a statement that does not take into account the existence of a probability. From

a probabilistic point of view, if we have a property that depends on sample points then we should agree that it holds provided those points where it fails has probability zero. This is what we discuss now.

Let $(\Omega, \mathcal{A}, P)$ be a probability space. For two random variables X and Y, say that $X \leq Y$ *a.e.* if $P[\omega : X(\omega) > Y(\omega)] = 0$. Say that $X = Y$ *a.e.* if $P[\omega : X(\omega) \neq Y(\omega)] = 0$. Say that $X_n \uparrow$ *a.e.* if $P[\omega : X_n(\omega) > X_{n+1}(\omega)$ for some $\omega] = 0$. Say that $X_n \to X$ *a.e.* if $P[\omega : X_n(\omega) \not\to X(\omega)] = 0$. Here *a.e.* is abbreviation for almost everywhere. Observe that to make sense of these definitions, we must make sure that the sets in braces are indeed events. But once you recognize the problem, it is easy to sort out. Most of the theorems on integration can be fine tuned by putting *a.e.* For example, MCT can be restated as follows : if $X_n \uparrow X$ *a.e.* and $X_n \geq 0$ *a.e.* then $E(X_n) \uparrow E(X)$. D.C.T can be restated as follows : If $X_n \in L^1$ for each n; $X_n \to X$ *a.e.* ; there is $Y \in L^1$ such that $|X_n| \leq Y$ *a.e.* for all n, then $X \in L^1$ and $\int |X_n - X| dP \to 0$ and $\int X_n dP \to \int X dP$.

5. Product spaces

Suppose that you have one model, that is, we have a probability space $(\Omega, \mathcal{A}, P)$. How to make new models from this.

Here is one way : Take one sub σ-field $\mathcal{B}$ of $\mathcal{A}$ and restrict your probability P to $\mathcal{B}$ and denote this by Q. Here sub σ-field means that $\mathcal{B}$ is a σ-field of subsets of Ω and every set which is in $\mathcal{B}$ is in $\mathcal{A}$. Then $(\Omega, \mathcal{B}, Q)$ is again a probability space. This will be useful if you already have partial information about the outcome of the experiment – for example you may already know which event in $\mathcal{B}$ actually occurred. Specifically,I may make an observation on a standard normal variable and may anounce to you absolute value of the outcome. There is unceratinty since you still do not know the sign of the observation. This set up will be useful to you when you define conditional expectation in a later course.

Here is another way : Pick a set $\Omega_0 \in \mathcal{A}$ and restrict your probability to this set and normalize. More precisely take $\mathcal{B} = \{B \subset \Omega_0 : B \in \mathcal{A}\}$. Define for sets B in this class $Q(B) = P(B)/P(\Omega_0)$. Of course for this to make sense we must assume that $P(\Omega_0) > 0$. I may perform the experiment and already tell you that the the observation is in Ω_0. There is no point in hanging on to the initial assumption that the chance of the observation falling in Ω_0 is $P(\Omega_0)$. We must modify our model to incorporate the given information and make $P(\Omega_0) = 1$. There are several reasons why this Q fits this job. This is conditional probability *given* Ω_0.

Here is another way : take a nonnegative random variable X with $E(X) = 1$ and put $Q(A) = \int X I_A \, dP$ for sets $A \in \mathcal{A}$. Then Q is a probability on $\mathcal{A}$ and thus $(\Omega, \mathcal{A}, Q)$ is a probability space. Here X is called density of Q w.r.t P. Thus normal density is actually density – the only trouble is that it is not density w.r.t another probability, but it is density w,r.t. the Lebesgue measure which is a σ-finite measure.

Here is yet another way : Suppose that we have a set Ω' and a σ-field $\mathcal{A}'$ of its subsets, and a map T defined on Ω with values in Ω' such that for every set $A' \in \mathcal{A}'$, we have $T^{-1}(A') \in \mathcal{A}$. Then we can define a probability on $\mathcal{A}'$ by putting $Q(A') = P(T^{-1}A')$. We get a new probability space $(\Omega', \mathcal{A}', Q)$. This

Q is called the induced probability on $\mathcal{A}'$ induced by T, or also the distribution of T. In particular if Ω' happens to be the real line with its Borel σ-field and T is a random variable X, then this Q is called the distribution of the random variable X. This arises in statistics quite often. The original space corresponds to the sample from a population and T is what is called *statistic* based on the sample.

We shall now describe another way of manufacturing new spaces out of old ones. This is a standard construction – product spaces. For two groups you can consider the product group, for vector spaces you can consider the product vector space etc. For probability spaces also we can construct product probability space. This is not just for the sake of imitation. This will be the main tool for getting independent random variables – or equivalently this models two independent experiments. This can aso be regarded as imitation of defining areas from lengths. Note that if we have a rectangle then its area is the product of lengths of its sides. Of course when we said length we had usual length in mind. But we can, if we wish, use different scales for the two axes. Why not?

Suppose we have two probability spaces $(\Omega_1, \mathcal{A}_1, P_1)$ and $(\Omega_2, \mathcal{A}_2, P_2)$. Define $\Omega = \Omega_1 \times \Omega_2$, that is, cartesian product of the two sets Ω_1 and Ω_2. For sets $A_1 \in \mathcal{A}_1$ and $A_2 \in \mathcal{A}_2$, consider the rectangle $A_1 \times A_2$. This is a subset of Ω. We do not discuss rectangles if sides are not in the respective σ-fields. The collection of all such rectangles as A_1 varies in $\mathcal{A}_1$ and A_2 varies in $\mathcal{A}_2$ is denoted by $\mathcal{R}$. Let $\mathcal{A} = \sigma(\mathcal{R})$. This is also denoted as $\mathcal{A}_1 \otimes \mathcal{A}_2$. Obviously for a rectangle $A_1 \times A_2$ we want its probability to be $P_1(A_1)P_2(A_2)$. Indeed there is a such a probability on the product space.

Theorem 11 : Product probability

There is a unique probability P on $\mathcal{A}$ such that for rectangles $P(A_1 \times A_2) = P_1(A_1)P_2(A_2)$.

This probability is called the product probability and is also denoted by $P_1 \otimes P_2$. Thus we have a new probability space

$$(\Omega, \mathcal{A}, P) = (\Omega_1 \times \Omega_2, \mathcal{A}_1 \otimes \mathcal{A}_2, P_1 \otimes P_2) = \otimes_{i=1}^{2} (\Omega_i, \mathcal{A}_i, P_i)$$

Of course it is necessary to explain how to calculate expectations w.r.t this new probability if we know how to do with the given spaces. In calculus, you evaluated double integrals using what is called repeated integration – or to put it differently, you explained how to integrate function of two variables if we know how to integrate functions of one variable. Of course to integrate one function of two variables, you need to integrate several functions of one variable. This is possible in our present situation too.

First some notation. Suppose that we have a function f on the product set Ω. For each $\omega_1 \in \Omega_1$ we denote by f_{ω_1} the function defined on Ω_2 by the formula $f_{\omega_1}(\omega_2) = f(\omega_1, \omega_2)$. This is called the vertical section of f at ω_1. Similarly we can define for each $\omega_2 \in \Omega_2$ a function f^{ω_2} on Ω_1 by the formula $f^{\omega_2}(\omega_1) = f(\omega_1, \omega_2)$. This is called the horizontal section of f at ω_2. Of course these are nothing but the same f with appropriate variable fixed.

Theorem 12 : Fubini

(i) Let X be a non-negative random variable on the product space. Then for each ω_1, X_{ω_1} is a random variable on Ω_2 and the map $\omega_1 \longrightarrow E_{P_2}(X_{\omega_1})$

is a random variable on Ω_1. Similarly for each ω_2, X^{ω_2} is a random variable on Ω_1 and the map $\omega_2 \longrightarrow E_{P_1}(X^{\omega_2})$ is a random variable on Ω_2. Finally $E_P(X) = E_{P_1}(E_{P_2}(X_{\omega_1})) = E_{P_2}(E_{P_1}(X_{\omega_2}))$.

(ii) Let X be a P- integrable random variable on the product space. Then for each ω_1, X_{ω_1} is a random variable on Ω_2 and also for *a.e.* , ω_1 it is P_2 integrable and the map $\omega_1 \longrightarrow E_{P_2}(X_{\omega_1})$ is a random variable on Ω_1 with the provision that for those ω_1 for which X_{ω_1} is not P_2 integrable we take the value to be zero. Similarly for each ω_2, X^{ω_2} is a random variable on Ω_1 and also for *a.e.* , every ω_2 it is P_1 integrable and the map $\omega_2 \longrightarrow E_{P_1}(X^{\omega_2})$ is a random variable on Ω_2 with the provision that for those ω_2 for which X_{ω_2} is not P_1 integrable we take the value to be zero. Finally $E_P(X) = E_{P_1}(E_{P_2}(X_{\omega_1})) = E_{P_2}(E_{P_1}(X_{\omega_2}))$.

The moral is that you can integrate the variables one by one as you did in calculus. The conclusion is also written as

$$\int X\, dP = \int (\int X_{\omega_1}\, dP_2)\, dP_1 = \int (\int X^{\omega_2}\, dP_1)\, dP_2$$

It should be noted that for nonnegative random variables we can always talk about expectation – at the worst it may turn out to be ∞. That is why we did not make fuss about integrability in the first part. For general random variables even the symbol $E(X)$ will not make sense unless you are sure that the random variable is integrable. This is one instance where you end up with extended random variables after one integration – remember your original X is real valued.

This theorem is very powerful and we shall see applications soon. The nuisance is that in the second part we demanded that X be P- integrable. You may wonder whether verification of this hypothesis is easy at all. If we need to verify this then whether we succeeded in explaining integartion w.r.t P using integrations w.r.t P_1 and P_2. You are right. The verification of the hypothesis does seem to need integartion w.r.t P. However the two parts of the theorem are to be taken together. Given a random variable X on the product space we consider first of all $|X|$ and apply part one. In part one there is no hypothesis. See whether $|X|$ is P-integrable by performing the two repeated integrals in your own convenient order – on $|X|$. Then if it is P-integrable then apply part two of the theorem to calculate $E_P(X)$ by performing the repeated integrals in your own convenient order – now on X.

Suppose we take μ to be the standard normal probability on the real line (with Borel σ-field). Consider the product space by taking each space to be $(R, \mathcal{B}, \mu)$. we get the product space $(R^2, \mathcal{B}^2, P)$ where $P = \mu \otimes \mu$. Now suppose that on the product space we consider the random variables : $X(a, b) = a$ and $Y(a, b) = b$ – that is, the co-ordinate random variables. These are indeed random variables. It is immediate that for any two numbers x and y ; $P(X \leq x, Y \leq y) = P(X \leq x)P(Y \leq y)$. In other words these random variables are independent. Further the distribution of each is standard normal – that is for each number x, $P(X \leq x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{x} exp(-t^2/2)\, dt$. Thus we have two independent standard normal variables.

When we built probability models and made our definition of distribution and random variables it was not clear whether we could manufacture two independent random variables. But as seen above, we can. Nothing special about standard normal. You can take any two distributions you want. Again there is

nothing special about two, you can manufacture any number. The concept of product can be extended to many spaces.

6. Inequalities

There are some useful inequalities that everyone ought to learn. First question is why should we learn inequalities. Here is one reason. In many instances it is not easy to calculate probabilities exactly. There are two possibilities then open for us. Either give a complicated expression as an answer, the answer may not reveal how large or how small the probability is. Or try to get a not so exact answer but some approximate answer which gives a better idea of the probability we are trying to calculate. For example toss a fair coin 10000 times. What is the probability that the number of heads is between 4500 and 5500. Of course we all learnt binomial distribution and can write the exact expression for its probability — as a sum of binomial probabilities. What good is it? If you try to calculate each term and add up, then they are so small that you end up with nearly zero upto several decimal places. But this does not mean that the sum is zero.

Here is another reason. We may not be interested in the exact value but only a bound for it. This happens especially when you discuss limits. This is not surprising. If you remember how you proved, in your calculus courses, convergence of sequences and series you will know. Then you may ask why should I care about limits. This question is easily answered. Suppose you are trying to estimate a parameter of a population, as we do in statistics. You take a sample and provide an estimate. It is necessary to know whether the estimate equals the parameter value when the sample equals the population. What does this mean? There are several ways of giving a meaning to this. The simplest way is to interpret whether the estimate converges to the true parameter as the sample size increases (somehow we believe that when the sample size increases, though it may never equal the population, it represents all the characterstics of the population).

At times you would also want to know how close are two random variables X and Y. If you are trying to understand the random variable X and if it turns out to be too complicated, but on the other hand Y is easy to handle then you would like to analyze the later random variable provided it is close to X and expect that X also would have similar properties. (Most of the tests of convergence of series of real numbers is via comparing with a well behaved simple series. Is it not?)

In a sense inequalities are really one of the main tools for a probabilist and statistician. It so happens that they are mathematically useful too. Here are some inequalities. Hopefully we see some applications later. A probability space $(\Omega, \mathcal{A}, P)$ is fixed in what follows.

Theorem 13 : Basic inequalities

(i) **Markov**'s inequality : For a non-negative random variable X and any $a > 0$, $P(|X| > a) \leq E|X|/a$.

(ii) **Tchebycheff**'s inequality : For any random variable X with mean μ and finite variance σ^2 , $P(|X - \mu| > a) \leq \sigma^2/a^2$.

(iii) **Kolmogorov**'s inequality : If $(X_i)_{1\leq i\leq n}$ are independent mean zero random variables and if S_i denotes their partial sum sequence, then for any $\epsilon > 0$, $P(\max_{1\leq i\leq n} |S_i| \geq \epsilon) \leq \frac{1}{\epsilon^2} \sum_1^n var(X_i)$.

(iv) **Jensen**'s inequality : X is an integrable random variable taking values in the interval (a, b) where $-\infty \leq a < b \leq \infty$, and φ is a convex function on the interval (a, b), and φoX is also integrable then $\varphi(EX) \leq E(\varphi oX)$.

(v) **Holder**'s inequality : For nonnegative random variables X and Y, and for numbers $p > 1$, $q > 1$ with $\frac{1}{p} + \frac{1}{q} = 1$; $E(XY) \leq (EX^p)^{1/p}(EY^q)^{1/q}$.

(vi) **Cauchy-Schwarz** inequality : The above inequality when $p = 2$ and $q = 2$ – that is, for random variables X and Y; $E|XY| \leq \sqrt{EX^2}\sqrt{EY^2}$

(vii) **Minkowski** inequality : For non-negative random variables X and Y, and $p \geq 1$, $(E(X + Y)^p)^{1/p} \leq (EX^p)^{1/p} + (EY^p)^{1/p}$.

(viii) **Borel-Cantelli** lemma : If A_n are events with $\sum P(A_n) < \infty$ then $P(limsup\, A_n) = 0$. If the events are independent and if $\sum P(A_n) = \infty$ then $P(limsup\, A_n) = 1$.

The Borel Cantelli lemma is one of the simplest and most powerful tool in probability.

7. Convergence concepts

Let $(\Omega, \mathcal{A}, P)$ be a probability space. Let (X_n) be a sequence of random variables and X be a random variable.

Say that $X_n \to X$ *everywhere* if for every sample point ω, $X_n(\omega) \to X(\omega)$. This concept has no reference to the underlying probability and could have been defined for functions on any set.

Here is a better concept that takes probability into account. Say that $X_n \to X$ *a.e.* if the set of sample points ω for which $X_n(\omega) \to X(\omega)$ has probability one. This concept involves collecting all sample points ω for which the sequence of numbers $X_n(\omega)$ converges to $X(\omega)$ and checking whether this event has probability one.

Here is a weaker concept you came across in WLLN. Say that $X_n \to X$ *in probability* if for every $\epsilon > 0$ we have $P(|X_n - X| > \epsilon) \to 0$. Here you need not actually collect sample points. If you knew the joint distribution of X_n and X, you can calculate the probability above for each ϵ and see if it converges to zero. Here the chances of X_n and X differing by an amount larger than a *preassigned* quantity goes to zero. On the face of it appears as if here also you are actually collecting sample points ω for which $|X_n(\omega) - X(\omega)| > \epsilon$ and calculating its probability (whether you do it on the sample space or using joint distribution). Yes, you are right. But you see this involves only two random variables, namely X_n and X. If you look at the *a.e* concept above, to see whether a sample point ω is to be collected or not depends on whether $X_n(\omega)$ converges to $X(\omega)$ or not. This involves looking at *all* your random variables at this point ω.

Here is a further weaker concept. Say that $X_n \to X$ *in law* or *in distribution* if $F_n(a) \to F(a)$ for each number a where F is continuous. Here F_n is the d.f of X_n and F is the distribution of X. Equivalently, $P(X_n \leq a) \to P(X \leq a)$ for each number a with $P(X = a) = 0$. Here the entire operation is shifted to the real line via distribution functions. This looks a little awkward because we

are demanding the convergence of distribution functions *only* at the continuity points of the limiting F. The reason is simple. Suppose $X_n \equiv 1/n$ and $X \equiv 0$. No matter what notion of convergence is, we expect X_n to converge to X. If you demanded the convergence of distribution functions at *every* point a then this does not happen in this example. You came across this concept in CLT. One interesting thing about this concept is that your random variables need not even be defined on the same space.

Here are yet other very useful concepts. Fix $1 \leq p < \infty$. Say that $X_n \to X$ *in* L^p if $||X_n - X||_p \to 0$ where for any random variable Z, $||Z||_p = (E|Z|^p)^{1/p}$. This is called L_p convergence or convergence in the p-th mean.

The first question that naturally arises is: why so many concepts? It depends on the problem at hand. We will have occassion to get into these issues as we go along. Next question is the relation between these concepts and their uses. Here are some basic results that explain the relations.

Theorem 14 : Relations between convergence notions

Let $(\Omega, \mathcal{A}, P)$ be a probability space.

(i) $X_n \to X$ a.e. implies $X_n \to X$ in probability, but the converse is not always true.

(ii) $X_n \to X$ in probability implies that $X_n \to X$ in distribution, but the converse is not always true even if all your random variables are defined on the same probability space.

(iii) Let $1 \leq p < \infty$. If $X_n \to X$ in L^p then $X_n \to X$ in probability, but the converse need not be true even if all the random variables are in the L^p we are talking about.

Theorem 15 : L^p Spaces

Completeness and separability $1 \leq p \leq \infty$ and also convergence in probability.

We shall now discuss some results which are not only interesting, but also lead to the law of large numbers.

Theorem 16 :Series of independent random variables

(i) If X_n are independent mean zero r.v.s, with $\sum Var(X_n) < \infty$ then the series $\sum X_n$ converges a.e. If the random variables are moreover bounded (besides being independent and mean zero) then the converse is also true – that is, if $\sum X_n$ converges a.e., then $\sum Var(X_n) < \infty$.

(ii) **Kolmogorov**'s three series theorem : Suppose that X_n are independent random variables. If there is a number $c > 0$ such that denoting $A_n = (\omega : |X_n(\omega)| \leq c)$ the following three numerical series converge:

$\sum P(A_n^c)$; $\sum \int_{A_n} X_n \, dP$ and $\sum [\int_{A_n} X_n^2 \, dP - (\int_{A_n} X_n \, dP)^2]$

Then $\sum X_n$ converges a.e. Conversely, if the series $\sum X_n$ converges a.e, then for *any* number $c > 0$, the above three numerical series converge.

Here is the Strong Law of Large Numbers.

Theorem 17: SLLN

(i) If X_n are independent mean zero r.v.s, with $\sum \frac{1}{n^2} Var(X_n) < \infty$ then $\frac{1}{n} \sum_1^n X_i \to 0$ a.e.

(ii) If X_n are i.i.d random variables with mean m then $\frac{1}{n} \sum_1^n X_i \to m$ a.e.

The SLLN is one of the fundamental theorems of probability for good reasons. Here are some.

We built a mathematical model for infinite coin tossing experiment – remember 2^ω and the coin tossing probability where the chance of heads in a single toss is p. When we had this p in mind, our intuitive feeling is that when you toss the coin a large number of times the proportion of heads we see is approximately p. The question is whether there is anything in our model to reflect this feeling of ours. If there is no such, then perhaps our model is good mathematics alright but is not the model we are looking for! Luckily, there is no problem. If X_n denotes the n-th coordinate on the coin tossing space then (X_n) is a sequence of independent random variables with common mean p. X_n takes the value one when the n-th toss results in heads. Thus S_n/n denotes the proportion of heads in the first n tosses. SLLN tells us that that this proportion indeed converges to p. This in a sense vindicates that the model is achieving what we are trying to do.

Here is a statistical reason. Suppose we do have a coin and do not know the chance of heads. We want to find out. Common sense tells us that we should toss the coin a large number times and propose the proportion of heads obtained as a reasonable guess for the value of p. The question then arises whether this does give the true unknown value if we really tossed the coin *forever*. The answer is provided by the SLLN. It says that almost surely, your guess does approach the true value. Of course, when you are dealing with chance phenomena, you should be satisfied with such almost sure statements.

The SLLN is satisfactory for other reasons too, which do not concern here, but we ought to know. Physicists are very clever who try to explain phenomena that we see — by using a few basic rules. If you blow a ballon and hold it carefully, it does not shrink immediately. The reason is that even though the air outside is pressing it, the air inside is also exerting a pressure on the balloon. What causes the pressure inside. It is the air molecules, which are constantly in motion, bombarding the walls of the balloon. So far so good. But then they go on to measure the pressure – the force exerted by the air molecules on unit area. But if the pressure is caused by the constant bombardment of the balloon by the air molecules it is reasonable to ask whether they are all moving with the same speed, whether all are exerting the same force when they hit and so on. Without any reasonable answer to these doubts, one wonders if the concept of pressure is well defined at all. But in practice you do see the balloon holding on well and not showing any appreciable changes (in a small time period at least). The main argument is that there are so many molecules in the balloon, they produce an average effect. It is this average effect that we call pressure.

There are other situations too. For example, in brain modelling also the 'average effect' phenomenon is essential. For instance you are trying to identify a person by looking at a given old and possible unclear picture. Obviously there are several differences between the picture you see and the person — of course each of these differences are pointed out to you by the neurons. In practice we do arrive at a decision inspite of receiving contradictory signals. This is probably because of the average effect produced.

Suppose that we have a probability space $(\Omega, \mathcal{A}, P)$ and for $n \geq 1$ $\mathcal{A}_n$ be sub σ-fields of $\mathcal{A}$. We say that these sub σ-fields are independent if for any $n \geq 1$

and for any sets $A_i \in \mathcal{A}_i$ for $1 \leq i \leq n$ we have $P(\cap_i^n A_i) = \prod_i^n P(A_i)$.

Theorem 18 : Kolmogorov 0 − 1 Law

Let $(\mathcal{A}_n)$ be a sequence of independent sub σ-fields of $\mathcal{A}$ in a space $(\Omega, \mathcal{A}, P)$. Let

$$\mathcal{T} = \cap_n \sigma(\mathcal{A}_n, \mathcal{A}_{n+1}, \cdots)$$

Then P is trivial on $\mathcal{T}$ – that is if $A \in \mathcal{T}$, then $P(A)$ is either zero or one.

This σ-field $\mathcal{T}$ is called the tail σ-field and sets in this are called tail sets – of course, everything is relative to the given sequence of independent σ-fields. This theorem says that for events in the tail σ-field there is a dichotomy – the probability of the event is either one or zero, nothing in between. In other words there is a typical behaviour of the sample points. To be specific suppose you have a series $\sum X_n$ of independent random variables. The event consisting of those sample points ω such that $\sum X_n(\omega)$ converges is a tail event. So this has probability one or zero. Thus the series exhibits a typical behaviour.

8. Characterstic functions

Suppose that X is a random variable defined on a probability space. Recall that $E(e^{tX})$ is called the moment generating function (m.g.f) of the random variable X. This function of t, reveals lot of information about the random variable. For example moments of the r.v. can be obtained from the m.g.f. Further, equality of the m.g.f s implies that the r.v.s have the same distribution. But the trouble is that the required expectation may not exist and the function may not be defined for any value of t. On the other hand if we define $\varphi(t) = E(e^{itX})$, then this exists for every real t. This is called the characterstic function (c.f. for short) of the random variable X. Note that this $\varphi(t)$ is nothing but $E(cos\, tX) + iE(sin\, tX)$. Actually, it is immaterial where your random variable is defined, what matters is the distribution of the r.v. In other words if X and Y are two r.v.s having the same distribution then they have the same c.f. If we denote the distribution of X by μ and the distribution function by F, then change of variable formula says that $\varphi(t) = \int e^{itx}\, d\mu(x)$. We shall also use the terminology that φ is the c.f. of μ or the c.f. of F. By a c.f. means c.f. of some probability μ.

Here are some basic results about the c.f.s

Theorem 19: Properties of Characterstic functions

(i) If φ is a c.f., then $\varphi(0) = 1$; for all t, $\varphi(-t) = \overline{\varphi(t)}$; φ is continuous; for all t, $|\varphi(t)| \leq 1$ and it is nonnegative definite – which means that for any n and any $t_1 < t_2 < \cdots t_n$, the matrix $(\varphi(t_i - t_j))_{1 \leq i,j \leq n}$ is nonnegative definite.

(ii) **Bochner**'s characterization : If you ever see a complex valued function on R satisfying the properties of (i) above, then be assured that it is a c.f. – that is, there is a probability μ on R whose c.f. is the function you see.

(iii) **Inversion formula :** If φ is the c.f. of F, then for any two numbers $a < b$, we have

$$\frac{F(b) + F(b-)}{2} - \frac{F(a) + F(a-)}{2} = \lim_{T \to \infty} \quad \frac{1}{2\pi} \int \frac{e^{-ibt} - e^{-iat}}{-it}\, \varphi(t)\, dt$$

In particular, if a and b are continuity points of F, then the rightside equals $F(b) - F(a)$.

(iv) **Uniqueness** theorem : Let μ and ν be two probabilities on R with c.f.s φ and ψ. If φ and ψ are same then the probabilities μ and ν are same. That is, if for every t, $\varphi(t) = \psi(t)$ then for every Borel set $A \subset R$, $\mu(A) = \nu(A)$.

(v) **Bochner**'s formula for jump : Let φ be c.f. of μ. For any $x \in R$

$$\mu\{x\} = \lim_{T\to\infty} \quad \frac{1}{2T} \int_{-T}^{+T} e^{-itx}\, \varphi(t)\, dt$$

(vi) **Wiener**'s criterion for continuity : μ is continuous (that is, F is continuous) iff $\lim_{T\to\infty} \frac{1}{T} \int_0^T |\varphi(t)|\, dt = 0$

(vii) If φ is the c.f. of X , $r > 1$ is an integer then $E(|X|^r) < \infty$ implies that φ is r times differentiable and indeed $\varphi^{(r)}(t) = \int (ix)^r\, e^{itx}\, d\mu(x)$. The converse need not be true. However $\varphi''(0)$ exists iff $E(X^2)$ exists.

(viii) **Levy - Cramer** continuity theorem : Let φ_n be the c.f. of F_n and φ be that of F. Then $F_n \Rightarrow F$ if and only if for every t, $\varphi_n(t) \to \varphi(t)$.

Here is a consequence of the Levy - Cramer continuity theorem, known as the Central Limit Theorem, which is one of the fundamental theorems of probability.

Theorem 20 Central Limit Theorem (CLT) :

Let X_n be i.i.d random variables with common mean θ and common variance σ^2 and denote by S_n the partial sum $X_1 + X_2 + \cdots + X_n$. Then $\frac{1}{n\sqrt{\sigma}}\,(S_n - n\theta) \Rightarrow N(0, 1)$.

9. Extras

Here are some very useful results whose proofs are generally excluded from elementary courses — not because they are complicated. Normally there is not enough time to cover these and also of course, they need more concentration than is usually possible in a first course.

Suppose that μ and ν are two probabilities on $(\Omega, \mathcal{A})$. Say that μ is absolutely continuous w.r.t ν, in symbols $\mu << \nu$, if $\nu(A) = 0$ implies $\mu(A) = 0$. Say that μ is singular w.r.t ν, in symbols $\mu \perp \nu$, if there is a set $A \in \mathcal{A}$ such that $\mu(A) = 0$ and $\nu(A^c) = 0$. Thus the two probabilities are concentrated on disjoint sets. This relation is symmetric. That is, if $\mu \perp \nu$ then $\nu \perp \mu$. Obviously absolute continuity is not symmetric. If both $\mu << \nu$ and $\nu << \mu$ hold, then we write $\mu \equiv \nu$.

Theorem 21 : Radon-Nikodym and Lebesgue Decomposition Theorems

(i) Let μ and ν be two probabilities on $(\Omega, \mathcal{A})$. Then $\mu << \nu$ iff there exists a random variable Z such that for all $A \in \mathcal{A}$, $\mu(A) = \int_A Z\, d\nu$. Further such a random variable Z is almost surely unique – that is, if Z_1 and Z_2 satisfy the condition then $\nu(Z_1 \neq Z_2) = 0$. Such a random variable is almost surely nonnegative.

(ii) Let μ and ν be two probabilities on $(\Omega, \mathcal{A})$. Then μ is sum of two uniquely defined measures μ_1 and μ_2 with $\mu_1 << \nu$ and $\mu_2 \perp \nu$.

Part (i) above is called the *Radon-Nikodym Theorem*. The random variable Z is called the *Radon-Nikodym derivative* of μ w.r.t ν — in symbols $Z = \frac{d\mu}{d\nu}$, one also writes $d\mu = Z\, d\nu$.

Part (ii) of the above theorem is called the *Lebesgue decomposition Theorem*. The measure μ_1 is called the *absolutely continuous part* of μ and μ_2 is called the *singular part* of μ. These are, of course, relative to ν. This theorem can also be restated as: μ is a convex combination of two probabilities; one absolutely continuous w.r.t ν and the other singular w.r.t ν.

A probability on $(R, \mathcal{B})$ is said to be *pure* if it is either discrete or absolutely continuous w.r.t the Lebesgue measure or continuous and singular w.r.t the Lebesgue measure. Here is a nice consequence of the zero -one law.

Theorem 22 : Jessen - Wintner Law
Let X_n be independent discrete random variables and $\sum X_n$ converges almost surely. Then law of $\sum X_n$ is pure.

Sometimes it is necessary to consider measures that are not necessarily positive. They are called signed measures. Here is the definition. Suppose we have a set Ω with a σ-field $\mathcal{A}$ on it. Let $\nu : \mathcal{A} \mapsto R$ such that it is countably additive, which means, as earlier whenever $\{A_n\}$ are disjoint sets in $\mathcal{A}$ then $\nu(\cup A_n) = \sum \nu(A_n)$. Union of sets does not depend on the order, whereas the sum of a series of number depends on the order. Thus the countably additivity implies that for any sequence of disjoint events A_n the series $\sum \nu(A_n)$ is absolutely convergent. A simple example of signed measure is the difference of two finite (positive) measures. In fact these are the only ones.

Theorem 23 : Jordan-Hahn decomposition
Given a signed measure ν there are two sets Ω^+ and Ω^- in $\mathcal{A}$ whose union is Ω such that every subset of Ω^+ which is in $\mathcal{A}$ has non-negative ν value and every subset of Ω^- which is in $\mathcal{A}$ has non-positive ν value.

Note that if we define $\nu^+(A) = \nu(\Omega^+ \cap A)$ and $\nu^-(A) = -\nu(\Omega^- \cap A)$ then these are two (positive) finite measures and clearly $\nu(A) = \nu^+(A) - \nu^-(A)$.

Note further that the sets asserted in the theorem are essentially unique. This means that if you have two other sets Ω^{++} and Ω^{--} verifying the theorem, then every subevent of $\Omega^+ \Delta \Omega^{++}$ has ν value zero and every subevent of $\Omega^- \Delta \Omega^{--}$ has ν value zero.

Sometimes one allows the value $+\infty$ or the value $-\infty$ for signed measures. Of course you can not allow both values at the same time.

We talked about finite products of measures. It is natural to enquire about infinite products. This is important for probabilists. Remember that products of probability spaces helped us to get independent random variables and infinite products help in getting sequences of independent random variables.

Recall that $\mathcal{B}$ is the Borel σ-field on R, the real line and $\mathcal{B}^n$ is the product σ-field on R^n. We can define, in a similar way, the product σ-field on R^∞, to

be denoted $\mathcal{B}^\infty$.

$$\mathcal{B}^\infty \quad = \quad \sigma\{B_1 \times B_2 \times B_3 \times \cdots : B_i \in \mathcal{B} \text{ for all } i\}$$

This is the σ-field on R^∞. Suppose that μ is a probability on R^∞. You can look at its marginal μ^n – sometimes also called projection – on the first n coordinates. This is defined as follows. Let π^n be the projection from R^∞ to R^n

$$\pi^n(x_1, x_2, \cdots) \quad = \quad (x_1, x_2, \cdots, x_n)$$

This is clearly (?) a measurable map. Now μ^n is nothing but the probability on R^n by π^n. More precisely

$$\mu^n(B) \quad = \quad \mu(\pi^n)^{-1}(B) \quad \text{for } B \in \mathcal{B}^n$$

Thus with the help of μ on R^∞, we have a sequence of probabilities μ^n. Of course, they are not on the same space. Instead μ^n sits on R^n. This sequence of probabilities are consistent. This means that if you take $m < n$ and consider the projection $\pi^{n,m} : R^n \mapsto R^m$ then the probability induced by μ^n on R^m is nothing but mu^m.

Theorem 24 : Kolmogorov Consistency Theorem
Given a consistent sequence of probabilities μ^n on R^n; there is a unique probability μ on R^∞ for which the first n-dimensional marginal is μ^n.

We conclude with one final notion that is more in the realm of set theory but has bearing in probability. If you know ordinals then there is nothing new here, we are talking about the first uncountable ordinal! If you do not know ordinals you need not worry because we need nothing complicated.

Consider the set $\omega = \{0, 1, 2 \cdots\}$ with the usual order. Then this set has the following properties: 1. If you take any $n \in \omega$ then the set of its predecessors $\{m \in \omega : m \leq n\}$ is finite. 2. The set ω itself is not finite. 3. Every non-empty subset of ω has a first element. In fact these properties characterize this set. Indeed if you have any set X with an order satisfying these properties then use property 3, to get its first element, name it 0. Consider the set minus zero, by property 2 it is non-empty, name its first element 1. Proceed this way. It does not end at any finite stage because of property 2. Does this exhaust all of our set X? Yes. If there is another element x then all our earlier points must be its predecessors and you contradict property 1. A word of caution is in order. So far we denoted the points of our sample spaces by ω. Unfortunately, here we chose to denote by ω the set of non-negative integers for some established good reasons which do not ceoncern us here.

Is there a set just one level higher? What does this mean? Just replace 'finite' above by 'countable'. More precisely, we are looking for a set X with a linear order such that: 1. If you take any α in your set X the set of its predecessors $\{\beta \in X : \beta \leq \alpha\}$ is countable. 2. The set X itself is not countable. 3. Every non-empty subset of X has a first element. The interesting point is that there are such sets (essentially unique again). Let us not try to construct such set, rather let us assume that we have such a set. Name it ω_1 and denote its elements by α, β, γ etc. As earlier, our set can not have a last element. If it has, then look at its set of predecessors and arrive at a contradiction.

Theorem 25 : No continuous probability on ω_1:

(Assume Axiom of choice) There is no probability defined on all subsets of ω_1 which gives zero mass to each singleton set.

This is a theorem of S.M. Ulam. A moment's reflection shows that this theorem leads to Theorem 1.

PART B :THE METHODS

We shall now go through the proofs of the theorems. Keep a pen and paper handy. Verify things as you go along.

Theorem 1 : This is not essential for what follows and is not used below. This follows from Theorem 25 which will be proved at the end.

Theorem 2 : This is an immediate consequence of Theorem 8 and will not be used till that is proved.

Theorem 3 : Consider all possible σ-fields on the set Ω that include the given class of sets $\mathcal{C}$. There is at least one such — namely the class of all subsets of Ω. Collect in a bag just those sets A which belong to all these σ-fields mentioned in the first sentence. This bag of sets is a σ-field, it includes the given class $\mathcal{C}$. This is what we are looking for.

Theorem 4 : A σ-field is a monotone class — in particular $\sigma(\mathcal{F})$ is a monotone class. Since $\mathcal{M}(\mathcal{F})$ is the smallest monotone class that includes $\mathcal{F}$ we conclude that $\mathcal{M}(\mathcal{F}) \subset \sigma(\mathcal{F})$ It remains to show that $\mathcal{M}(\mathcal{F}) \supset \sigma(\mathcal{F})$.

We now show that $\mathcal{M}(\mathcal{F})$ is a field. What does this achieve? Since a field which is a monotone class is a σ-field, this shows that $\mathcal{M}(\mathcal{F})$ is indeed a σ-field. It then follows that $\mathcal{M}(\mathcal{F})$ is a σ-field that includes the class $\mathcal{F}$, so must include the smallest such namely, $\sigma(\mathcal{F})$.

First, we show that whenever $A \in \mathcal{M}(\mathcal{F})$ then so is A^c. For this consider the class $\mathcal{M}_0 = \{A \in \mathcal{M}(\mathcal{F}) : A^c \in \mathcal{M}(\mathcal{F})\}$. If $A \in \mathcal{F}$, then $\mathcal{F}$ being a field we have $A^c \in \mathcal{F}$ which is included in $\mathcal{M}(\mathcal{F})$, it follows that $\mathcal{F} \subset \mathcal{M}_0$. If each A_n is in $\mathcal{M}_0$ and $A_n \uparrow A$, then firstly A_n as well as A_n^c are in $\mathcal{M}(\mathcal{F})$. Secondly, $A_n \uparrow A$ as well as $A_n^c \downarrow A^c$ so that both A and A^c are in $\mathcal{M}(\mathcal{F})$. Thus $A \in \mathcal{M}_0$. Similarly if each A_n is in $\mathcal{M}_0$ and $A_n \downarrow A$ then $A \in \mathcal{M}_0$. In other words $\mathcal{M}_0$ is a monotone class that includes $\mathcal{F}$. But $\mathcal{M}(\mathcal{F})$ is the smallest such. This means that every set in $\mathcal{M}(\mathcal{F})$ must be in $\mathcal{M}_0$. Thus, whenever $A \in \mathcal{M}(\mathcal{F})$ so is A^c.

Second, we show that if A and B are in $\mathcal{M}(\mathcal{F})$ then so is their intersection $A \cap B$. To start with observe that if both the sets A and B are in $\mathcal{F}$ then their intersection is already in $\mathcal{F}$ and hence in $\mathcal{M}(\mathcal{F})$. Fix $A \in \mathcal{F}$. Consider $\mathcal{M}_0 = \{B \in \mathcal{M}(\mathcal{F}) : A \cap B \in \mathcal{M}(\mathcal{F})\}$. Beacuse of the earlier sentence, this family includes $\mathcal{F}$. As in the earlier paragraph, $\mathcal{M}_0$ is a monotone class and hence must be $\mathcal{M}(\mathcal{F})$. In otherwords, for every $B \in \mathcal{M}(\mathcal{F})$ we have $A \cap B \in \mathcal{M}(\mathcal{F})$. Now fix any $B \in \mathcal{M}(\mathcal{F})$ and consider $\mathcal{M}_0 = \{A \in \mathcal{M}(\mathcal{F}) : A \cap B \in \mathcal{M}(\mathcal{F})\}$. From what we proved just now, this family includes $\mathcal{F}$. This is again a monotone class and hence must be $\mathcal{M}(\mathcal{F})$. Thus intersection of two sets in $\mathcal{M}(\mathcal{F})$ is again in $\mathcal{M}(\mathcal{F})$.

This completes the proof.

Theorem 5 : Define, $B_1 = A_1$, $B_2 = A_2 - A_1$, $\cdots$, $B_k = A_k - A_{k-1} \cdots$; These are disjoint with union A and hence $P(A) = \sum P(B_k)$. By definition of sum of series the right side is nothing but $\lim_n \sum_1^n P(B_k)$, but this finite sum is $P(A_n)$ to complete proof.

To prove the other part you can use $A_n^c \uparrow A^c$. Or can argue as follows. Suffices to show $P(A_n - A) \downarrow 0$. Observe that $A_1 - A = (A_1 - A_2) \cup (A_2 - A_3) \cup \cdots$ so that $P(A_1 - A) = \sum P(A_{n-1} - A_n)$. But then the tail sum of the series on the right side, which is precisely $P(A_n - A)$, goes to zero.

Remark : Suppose that P is a probability on $\mathcal{F}$, a field of subsets of Ω. Now, countable additivity means that, if $A \in \mathcal{F}$ is union of a sequence of pairwise disjoint sets A_n where each A_n is in $\mathcal{F}$, then $P(A) = \sum P(A_n)$. Then for any sequence of sets A_n in the field which increase to a set A in the field, we have $P(A_n) \uparrow P(A)$. The proof is just as given above. Similar result holds for decreasing sequence of sets in the field. This fact will be used in the proof of Theorem 7.

Theorem 6 : The class of all sets $A \in \mathcal{A}$ such that $P(A) = Q(A)$ includes the field by hypothesis, it is a monotone class by previous theorem and hence must equal $\mathcal{A}$, by MCT.

Theorem 7 : This is one of the few lengthy – but natural — proofs. In view of the earlier theorem there can not be two different extensions. We only need to exhibit one extension. We shall follow Jacque Neveu. This is carried out in three steps.

Step 1: Let

$$\mathcal{F}_\sigma = \{A : \text{ there is a sequence of sets } A_n \in \mathcal{F}; A_n \uparrow A\}.$$

Of course, since $\mathcal{F}$ is a field, this class is nothing but countable unions of sets in $\mathcal{F}$. Guided by theorem 6, for A in this class put $P(A) = \lim_n P(A_n)$. Since the sequence A_n is increasing, this limit exists. This is a good definition because if you take another sequence B_n from the field increasing to A, then for every m, $(B_n \cap A_m) \uparrow A_m$ and all these sets are in $\mathcal{F}$ so that by theorem 6 (really, by the remark following proof of theorem 6) $P(B_n \cap A_m) \uparrow P(A_m)$ and hence, for every m, $\lim_n P(B_n) \geq \lim_n P(B_n \cap A_m) = P(A_m)$. True for all m, implying $\lim_n P(B_n) \geq \lim_m P(A_m)$. Similarly $\lim_m P(A_m) \geq \lim_n P(B_n)$.

Note that if $A \in \mathcal{F}$ then we can take each A_n to be A itself to see that the value $P(A)$ defined in the earlier para is same as the starting value. Thus what we have defined is an extension of the given P. Thus using the same symbol P for this extension does not lead to any confusion. This extension P takes values in $[0, 1]$. It satisfies three properties that we need.

(1 a) It is monotone — that is, if $A \subset B$ then $P(A) \leq P(B)$. Indeed if $A_n \uparrow A$ and $B_n \uparrow B$ then $A_n \cap B_n$ is in $\mathcal{F}$ for each n and $(A_n \cap B_n) \uparrow A$ so that $P(A) = \lim P(A_n \cap B_n) \leq \lim P(B_n) = P(B)$.

(1 b) If $A_n \in \mathcal{F}_\sigma$ and $A_n \uparrow A$, then $P(A_n) \uparrow P(A)$. Indeed let $A_{m,n} \uparrow_m A_n$ for each n where all these sets $A_{m,n}$ are in $\mathcal{F}$. Put $B_n = \cup_{i,j \leq n} A_{i,j}$. Note that $B_n \uparrow A$ and all these sets B_n are in $\mathcal{F}$. This first of all shows that $A \in \mathcal{F}_\sigma$ and $P(A) = \lim P(B_n)$. Using the monotonicity of the sequence of sets A_n, observe

that $B_n \subset A_n$ so that $P(B_n) \leq P(A_n)$. Thus

$$P(A) = \lim P(B_n) \leq \lim P(A_n) \leq P(A)$$

the last inequality is from (a), because $A_n \subset A$ for all n. This shows that $P(A) = \lim P(A_n)$.

(1 c) If A and B are in $\mathcal{F}_\sigma$ then $P(A) + P(B) = P(A \cup B) + P(A \cap B)$. Indeed if $A_n \uparrow A$ and $B_n \uparrow B$ then $(A_n \cap B_n) \uparrow (A \cap B)$ and $(A_n \cup B_n) \uparrow (A \cup B)$ showing that both $A \cap B$ and $A \cup B$ are in $\mathcal{F}_\sigma$. Since the sets A_n, B_n, are all in $\mathcal{F}$ we also have

$$P(A_n) + P(B_n) = P(A_n \cup B_n) + P(A_n \cap B_n)$$

Taking limits and using the definition of our extension we get the stated equality.

Step 2: For every subset $B \subset \Omega$ let us put

$$P^*(B) = \inf\{P(A) : B \subset A \in \mathcal{F}_\sigma\}.$$

If at all we want to extend our probability this is the maximum value we can give for the set B. Clearly this P^* also takes values in the interval $[0, 1]$.

If B is already in $\mathcal{F}_\sigma$ then note that B itself is one candidate for the set A in the definition and any other A would have a possibly larger value by (1 a), showing that $P^*(B) = P(B)$. Thus P^* extends P from $\mathcal{F}_\sigma$. This P^* defined on all subsets of Ω has three properties that we need.

(2 a) It is monotone — that is, if $B_1 \subset B_2$, then $P^*(B_1) \leq P^*(B_2)$. This is immediate because any set that includes B_2 includes B_1 as well.

(2 b) For any two sets B_1 and B_2;

$$P^*(B_1) + P^*(B_2) \geq P^*(B_1 \cup B_2) + P^*(B_1 \cap B_2)$$

To see this, fix $\epsilon > 0$, pick for $i = 1, 2$; $A_i \supset B_i$, A_i in $\mathcal{F}_\sigma$ such that $P(A_i) \leq P^*(B_i) + \epsilon$. Then

$$P^*(B_1) + P^*(B_2) + 2\epsilon \geq P(A_1) + P(A_2) = P(A_1 \cup A_2) + P(A_1 \cap A_2)$$

The equality above is a consequence of (1c). But since $A_1 \cup A_2 \supset B_1 \cup B_2$ and $A_1 \cup A_2 \in \mathcal{F}_\sigma$ we deduce that $P(A_1 \cup A_2) \geq P^*(B_1 \cup B_2)$ and similarly $P(A_1 \cap A_2) \geq P^*(B_1 \cap B_2)$. Using this in the display above, we get

$$P^*(B_1) + P^*(B_2) + 2\epsilon \geq P^*(B_1 \cup B_2) + P^*(B_1 \cap B_2)$$

Since ϵ is arbitrary, the statement is verified.

(2 c) If $B_n \uparrow B$ then $P^*(B_n) \uparrow P^*(B)$. Of course by (2a) we know that $P^*(B_n)$ is increasing and is atmost $P^*(B)$. Thus $\lim_n P^*(B_n) \leq P^*(B)$. Towards proving the reverse inequality, fix $\epsilon > 0$. For each $n \geq 1$, fix $A_n \supset B_n$, $A_n \in \mathcal{F}_\sigma$

and $P(A_n) \leq P^*(B_n) + \epsilon/2^n$. Set $C_n = \cup_{m \leq n} A_m$ so that C_n are increasing, $C_n \in \mathcal{F}_\sigma$, $C_n \supset B_n$. We claim that for each n,

$$P(C_n) \leq P^*(B_n) + \sum_{m \leq n} \frac{\epsilon}{2^m}$$

This is clear, by choice of A_1, for $n = 1$. Going by induction, suppose we did upto n. Using (1c),

$$P(C_{n+1}) = P(C_n \cup A_{n+1}) = P(C_n) + P(A_{n+1}) - P(C_n \cap A_{n+1})$$

Now induction hypothesis and choice of A_{n+1} gives

$$P(C_{n+1}) \leq P^*(B_n) + \sum_{m \leq n} \frac{\epsilon}{2^m} + P^*(B_{n+1}) + \frac{\epsilon}{2^{n+1}} - P(C_n \cap A_{n+1})$$

Since $C_n \supset A_n \supset B_n$ and $A_{n+1} \supset B_{n+1} \supset B_n$ we see that $C_n \cap A_{n+1}$ is a set in $\mathcal{F}_\sigma$ which includes B_n so that $P(C_n \cap A_{n+1}) \geq P^*(B_n)$. Using this in the display above we get

$$P(C_{n+1}) = P^*(B_{n+1}) + \sum_{m \leq n+1} \frac{\epsilon}{2^m}$$

to complete the inductive step.

If $C = \cup_n C_n$, then $C_n \uparrow C$. By (1b), $C \in \mathcal{F}_\sigma$,

$$P(C) = \lim_n P(C_n) \leq \lim_n P^*(B_n) + \epsilon$$

from what we observed above. Since $C \supset B$, this implies

$$P^*(B) \leq P(C) \leq \lim_n P^*(B_n) + \epsilon$$

and ϵ being arbitrary we get $P^*(B) \leq \lim_n P^*(B_n)$ completing the proof.

Step 3 : It is natural now to get a lower bound for the value $P(B)$ we can assign to B and then collect those sets for which both the upper and lower bounds coincide – because for such sets there is nothing we should decide! But a moment's reflection shows that the lower bound for the value of $P(B)$ is nothing but $1 - P^*(B^c)$. Thus the condition that both uper and lower bounds coincide is the same as saying $P^*(B) + P^*(B^c) = 1$. Let $\mathcal{B}$ be the collection of all sets B satisfying this condition, that is, $P^*(B) + P^*(B^c) = 1$. We now study properties of this class $\mathcal{B}$ and properties of the restriction of P^* to this class.

(3 a) We first show that $\mathcal{B}$ is a field and P^* on this class is additive.

By the very definition of the class $\mathcal{B}$ it is clear that if B is in this class then so is B^c and hence $\mathcal{B}$ is closed under complements. To show $\mathcal{B}$ is closed under finite unions, take A and B in $\mathcal{B}$. So,

$$P^*(A) + P^*(A^c) = 1; \quad P^*(B) + P^*(B^c) = 1 \tag{1}$$

Apply (2 b) to A and B,

$$P^*(A) + P^*(B) \geq P^*(A \cup B) + P^*(A \cap B) \tag{2}$$

Apply (2 b) to A^c and B^c,

$$P^*(A^c) + P^*(B^c) \geq P^*((A \cup B)^c) + P^*((A \cap B)^c) \tag{3}$$

Add (2) and (3) and use (1)

$$2 \geq P^*(A \cup B) + P^*(A \cap B) + P^*((A \cup B)^c) + P^*((A \cap B)^c) \tag{4}$$

Apply (2b) to $A \cup B$ and $(A \cup B)^c$

$$P^*(A \cup B) + P^*((A \cup B)^c) \geq 1 \tag{5}$$

Apply (2b) to $A \cap B$ and $(A \cap B)^c$

$$P^*(A \cap B) + P^*((A \cap B)^c) \geq 1 \tag{6}$$

Strict inequality in either (5) or (6) will $--$ after adding (5) and (6) $--$ contradict (4). Hence equality must hold in both (5) and (6), which means that both $A \cup B$ and $A \cap B$ are in $\mathcal{B}$.

Note that equality in (5) and (6) deduced above implies that equality holds in (4) as well. Since strict inequality in either (2) or (3) will again lead to strict inequality in (4), we conclude that equality holds in (2) and (3) as well. In particular, we have

$$P^*(A) + P^*(B) = P^*(A \cup B) + P^*(A \cap B)$$

All this amounts to saying that $\mathcal{B}$ is a field and P^* is finitely additive on this class $--$ as claimed.

(3b) We now show that $\mathcal{B}$ is a σ-field and P^* is countably additive on this. Since it is already finitely additive, it suffices to take $B_n \uparrow B$ with each $B_n \in \mathcal{B}$ and show that $B \in \mathcal{B}$ and $P^*(B_n) \uparrow P^*(B)$. But this last statement is already (2c), so we only need to show $B \in \mathcal{B}$, that is $P^*(B) + P^*(B^c) = 1$. Of course, by (2b), $P^*(B) + P^*(B^c) \geq 1$. So we only need to show $P^*(B) + P^*(B^c) \leq 1$. Now, since each $B_n \in \mathcal{B}$ we have $P^*(B_n) + P^*(B_n^c) = 1$. But $B_n \subset B$ so that $B_n^c \supset B^c$. Consequently, $P^*(B^c) \leq P^*(B_n^c)$ Thus $P^*(B_n) + P^*(B^c) \leq 1$. Take limit over n and use (2c) to get the desired inequality.

(3c) Since P is a probability on $\mathcal{F}$; for $B \in \mathcal{F}$ we have $P(B) + P(B^c) = 1$. Since P^* extends P, the same holds for P^* as well. In other words, $\mathcal{F} \subset \mathcal{B}$. Now (b) above implies that $\sigma(\mathcal{F}) \subset \mathcal{B}$ and P^* is a probability on this which extends the given P on $\mathcal{F}$. To complete the proof of the theorem, we only need to look at P^* on $\sigma(\mathcal{F})$.

Theorem 8 : This is an application of the Caratheodory extension theorem. Read the statement of the theorem to convince yourself that you have no alternative but to define $P(-\infty, b] = F(b)$. Since you are searching for a probability, you are forced to put $P(a, b] = F(b) - F(a)$ for any $-\infty \leq a \leq b \leq \infty$, with the understanding $F(+\infty) = 1$ and $F(-\infty) = 0$. So consider the collection $\mathcal{S}$ of sets of the form $(a, b]$ with $-\infty \leq a \leq b \leq +\infty$. The convention is that $(a, \infty]$

is interpreted as (a, ∞). Thus for example $(-\infty, +\infty] = R$. We can write the proof without such conventions, but you have to consider cases. Define P for sets in the class as suggested above.

Step 1: We claim that P so defined is countably additive on the class of sets $\mathcal{S}$ — that is, if $I \in \mathcal{S}$ is a countable disjoint union of sets $I_n \in \mathcal{S}$ then $P(I) = \sum P(I_n)$. This will be done in a series of three observations.

1°. If we have a finite number of disjoint intervals $(a_n, b_n]$ for $1 \leq n \leq k$, all contained in $(a, b]$ then $F(b) - F(a) \geq \sum[F(b_n) - F(a_n)]$. Indeed, since we have only finitely many intervals, rearrange them if necessary and assume $a_1 < a_2 < a_3 < \cdots a_k$. Keep in mind the telescopic sum

$$F(b) - F(a) = [F(b) - F(b_k)] + [F(b_k) - F(a_k)] + [F(a_k) - F(b_{k-1})] + \cdots$$

and ignore the unnecessary terms – all terms being positive – you get the desired inequality.

2°. If we have intervals (a_n, b_n) for $1 \leq n \leq k$, whose union includes $[a, b]$ then $F(b) - F(a) \leq \sum_1^k [F(b_n) - F(a_n)]$. Indeed this is clearly true for $k = 1$ because F is non-decreasing. Suppose it is true for some k. If you have $k + 1$ intervals take the interval that includes the point b – without loss of generality assume it to be the $(k + 1)$-st interval. Observe that the remaining k intervals cover $[a, a_{k+1}]$. Use induction hypothesis and add one more term and simplify.

3°. Finally, let $I_n = (a_n, b_n]$ be disjoint with union $I = (a, b]$. Need to show that $F(b) - F(a) = \sum[F(b_n) - F(a_n)]$. By observation 1^0, the inequality $F(b) - F(a) \geq \sum[F(b_n) - F(a_n)]$ holds for each finite sum on the right side and hence holds for the total sum. To get the reverse inequality, fix $\epsilon > 0$. Choose $a' \in (a, b]$ with $F(a') < F(a) + \epsilon$. Even if $a = -\infty$, note that a' is finite. In case $b = \infty$, choose $b' \in (a, b)$ with $b' \geq a'$ so that $F(b') > F(\infty) - \epsilon = 1 - \epsilon$. If $b \neq \infty$ just take $b' = b$. For each n, choose $b'_n > b_n$ so that $F(b'_n) < F(b_n) + \epsilon/2^n$. Of course if $b_n = \infty$ just take $b'_n = b_n$. Clearly the intervals (a_n, b'_n) cover the compact interval $[a', b']$. Take a finitely many of these intervals that cover $[a', b']$. Apply observation 2^0 to see

$$F(b') - F(a') \leq \sum [F(b'_n) - F(a_n)] \leq \sum [F(b_n) - F(a_n)] + \epsilon$$

But in turn, $F(b) - F(a) \leq F(b') - F(a') + 2\epsilon$. Since $\epsilon > 0$ is arbitrary desired inequality is established.

Step 2: This collection of sets $\mathcal{S}$ has three interesting properties ; (1) $\emptyset, R \in \mathcal{S}$. (2) $A, B \in \mathcal{S}$ implies $A \cap B \in \mathcal{S}$. (3) If $A \in \mathcal{S}$ then A^c is a finite disjoint union of sets in $\mathcal{S}$. Such a collection of sets is called a *semifield* of subsets of R. What you name it is unnecessary for us now. Let the collection of finite disjoint unions of sets in this class be denoted by $\mathcal{F}$. Using the three above properties of $\mathcal{S}$ we verify that $\mathcal{F}$ is a field.

Define for $A \in \mathcal{F}$ which is a disjoint union of finitely many sets $A_n \in \mathcal{S}$, $P(A) = \sum P(A_n)$. This is a good definition because if the same A is a disjoint

union of finitely many sets B_m, $1 \le m \le l$ from $\mathcal{S}$ then

$$\sum_n P(A_n) = \sum_n \sum_m P(A_n \cap B_m) = \sum_m \sum_n P(A_n \cap B_m) = \sum_m P(B_m)$$

Here the first equality is from the facts that for each n, A_n is the union of disjoint sets $\{A_n \cap B_m : 1 \le m \le l\}$, these are all in $\mathcal{S}$, and by step 1, P is additive on this class; second equality is just interchange of the order of sums; and the third equality is from the fact that for each m, B_m is the union (now over n) of the disjoint sets $A_n \cap B_m$.

This P is actually countably additive on $\mathcal{F}$. Indeed let A_n be disjoint sets from $\mathcal{F}$ whose union, say, A is again in $\mathcal{F}$. Each of these sets in turn are disjoint union of finitely many sets from $\mathcal{S}$, say $A_n = \cup_j B_{n,j}$ and $A = \cup B_i$. Of course for different n, you may need different number of sets – that is the range of j in the union may depend on n. Let it be so. Now, using additivity of P from step 1 and the fact that intersection of two sets in $\mathcal{S}$ is again in $\mathcal{S}$, we get

$$\begin{aligned}
P(A) &= \sum_i P(B_i) && \text{by definition of } P(A) \\
&= \sum_i \sum_n P(B_i \cap A_n) && B_i \text{ is disjoint union of } B_i \cap A_n \\
&= \sum_i \sum_n \sum_j P(B_i \cap B_{n,j}) && A_n \text{ is disjoint union of } B_{n,j} \\
&= \sum_n \sum_i \sum_j P(B_i \cap B_{n,j}) && \text{interchange of order of sum} \\
&= \sum_n P(A_n) && A_n \text{ is dijoint union over } i,\ j \text{ of } B_i \cap B_{n,j}
\end{aligned}$$

To complete the proof appeal to Caratheodory and verify that $\sigma(\mathcal{F})$ is precisely the Borel σ-field of R.

Theorem 3B : Consider the collection of sets $X^{-1}(B)$ obtained by taking $X \in \Phi$ and Borel set $B \subset R$. Apply theorem 3 and check that the σ- field so obtained is what we want.

Theorem 9 : To prove X is a random variable it suffices to show that for any number a, the set $X^{-1}(-\infty, a)$ is in our σ-field, because then, $X^{-1}(-\infty, a] = \bigcap X^{-1}(-\infty, a + \frac{1}{n})$ will also be in our σ-field.

(i) For any number a, $(X + Y < a) = \cup_r \{(X < r) \cap (Y < a - r)\}$ where the union runs over the countable set of rationals r. Thus if X and Y are random variables then so is $X + Y$. For $\alpha > 0$, $(\alpha X < a) = (X < a/\alpha)$ and for $\alpha < 0$, $(\alpha X < a) = (X > a/\alpha)$ to conclude that αX is a random variable if X is so. Since $(X^2 \le a)$ is empty set if $a < 0$ where as it equals $(-\sqrt{a} \le X \le +\sqrt{a})$ for $a > 0$ showing that X^2 is a random variable if X is so. Thus in particular combining these it follows that if X and Y are random variables than so is $XY = [(X+Y)^2 - X^2 - Y^2]/2$. Observe that $(X \wedge Y < a) = (X < a) \cup (Y < a)$ and $(X \vee Y < a) = (X < a) \cap (Y < a)$ so that $X \wedge Y$ and $X \vee Y$ are random variables if X and Y are so. Since constant functions are visibly random variables this proves (i).

(ii) If $X_n \uparrow X$ then $(X \leq a) = \cap(X_n \leq a)$. If $X_n \downarrow X$ then $(X < a) = \cup(X_n < a)$ to prove the first statement. Let X_n be random variables. Fix n. For each $m > n$, $\max\{X_n, X_{n+1}, \cdots, X_m\}$ is a random variable by (i) and increases to, say, Y_n as $m \uparrow \infty$. By the first statement each Y_n is a random variable. But then $Y_n \downarrow \limsup X_n$ so that this last one is also a random variable. Similar argument holds for liminf.

(iii) Since $(I_A < 1) = A^c$ it follows that I_A is a random variable iff A is in our σ-field. For the second statement, you only need to verify that sums ,products, scalar multiples of simple functions are simple again.

(iv) For convenience assume that $0 \leq X \leq 1$. Define $s_n(\omega)$ as follows: if $0 \leq X(\omega) \leq 1/2^n$ then $s_n(\omega) = 0$ and for $1 \leq k \leq 2^n - 1$ put $s_n(\omega) = k/2^n$ in case $k/2^n < X(\omega) \leq (k+1)/2^n$. This sequence does.

Given non-negative X and an integer $n \geq 1$ define s_n as follows: $s_n(\omega) = 0$ in case $0 \leq X(\omega) \leq 1/2^n$ where $s_n(\omega) = n$ in case $X(\omega) > n$. For $1 \leq k \leq n2^n - 1$, in case $k/2^n < X(\omega) \leq (k+1)/2^n$ put $s_n(\omega) = k/2^n$. This sequence does.

(v) First observe that the class of sets A such that $I_A \in \Phi$ is a monotone class and includes $\mathcal{F}$ — all this by hypothesis. So by MCT, we conclude that $I_A \in \Phi$ for each $A \in \mathcal{A}$. Since Φ is a vector space, it must include every simple random variable. By hypothesis and (iv) it must include every nonnegative random variable too. For any random variable X, since $X = X^+ - X^-$ we conclude that X must also be in Φ to complete proof.

(vi) The required ideas are already in (v) above.

Theorem 10 :

(i) We shall show well-defined-ness shortly. Accepting this the other parts can be observed easily as follows. Suppose that $s = \sum_1^n a_i I_{A_i}$ and $t = \sum_1^m b_j I_{B_j}$ then clearly $s + t = \sum_1^{n+m} c_i I_{C_i}$ where $c_i = a_i$, $C_i = A_i$ for $1 \leq i \leq n$ and $c_i = b_{i-n}$, $C_i = B_{i-n}$ for $n+1 \leq i \leq n+m$. Now use the definitions of expectation to see $E(s+t) = E(s) + E(t)$.

(ii) Just to avoid confusion, temporarily use $\tilde{E}$ for expectation defined for non-negative random variables. Let X be simple nonnegative random variable. Note that X itself is a candidate for s in the definition of $\tilde{E}(X)$. Thus $\tilde{E}(X) \geq E(X)$. On the other hand if we take any other non-negative simple $s \leq X$ then monotonicity of expectation, observed in (i) above, yields that $E(s) \leq E(X)$ showing that $\tilde{E}(X) = E(X)$. Monotonicity of expectation is immediate from definition.

(iii) Start observing that if s is a simple nonnegative random variable and Ω_n are sets in the σ-field increasing to Ω then $E(sI_{\Omega_n}) \uparrow E(s)$. This is done by direct computation. For the proof of MCT, let $X_n \uparrow X$ – all being nonnegative. We already know that $E(X_n) \leq E(X)$ and increases with n , so that $limE(X_n) \leq E(X)$. Towards the other inequality fix any simple nonnegative $s \leq X$. Suffices to show that $E(s) \leq limE(X_n)$. Again fix $0 < \alpha < 1$. Suffices to show that $E(\alpha s) \leq limE(X_n)$. Let $\Omega_n = \{\omega : \alpha s(\omega) < X_n(\omega)\}$, so that $\Omega_n \uparrow \Omega$.

By monotonicity of expectation, we have $E(\alpha s I_{\Omega_n}) \leq E(X_n I_{\Omega_n}) \leq E(X_n)$. Take limits and use the observation made at the beginning to complete the proof.

(iv) Get simple nonnegative $s_n \uparrow X$ and $t_n \uparrow Y$. From (i), $E(s_n + t_n) = E(s_n) + E(t_n)$, use MCT for all the three terms noting that $s_n + t_n \uparrow X + Y$ to complete the proof.

(v) That the definition extends from non-negative varibles is clear since, if $X \geq 0$ then $X^+ = X$ and $X^- = 0$ so that the new definition is same as the old one. First note that X is integrable iff both $E(X^+)$ and $E(X^-)$ are finite – equivalently,$E(|X|) = E(X^+) + E(X^-)$ is finite. thus note that $X \in L^1$ iff $|X| \in L^1$. Further by definition, $|E(X)| = |E(X^+) - E(X^-)|$ is at most $E(X^+) + E(X^-)$ which is $E(|X|)$.

In particular, if X and Y are integrable, then so are $|X|$ and $|Y|$ and hence their sum because of additivity of expectation on non-negative random variables. Since integral is monotone on non-negative random variables and since $|X+Y| \leq |X| + |Y|$ we conclude that $E(|X + Y|)$ is finite and hence $X + Y$ is integarble. Finally, since

$$(X + Y)^+ - (X + Y)^- \quad = X + Y = \quad X^+ - X^- + Y^+ - Y^-$$

we conclude that

$$(X + Y)^+ + X^- + Y^- = (X + Y)^- + X^+ + Y^+$$

Now take expectation on both sides and since all terms are finite, rearrange them carefully to get $E(X + Y) = E(X) + E(Y)$. Monotonicity is routine.

(vi) If $Y_n = \inf\{X_n, X_{n+1}, \cdots\}$ then $Y_n \leq X_n$. So $E(Y_n) \leq E(X_n)$ for all n which implies that $\liminf E(Y_n) \leq \liminf E(X_n)$. Note that Y_n are increasing to $\liminf X_n$ and so by MCT, the left side is actually $E(\liminf X_n)$ to complete the proof.

(vii) The hypotheses imply that for the limit random variable X too we have $|X| \leq Y$ and hence X is integrable. So are $|X_n - X|$. The non-negative random variables $2Y - |X_n - X|$ converge pointwise to $2Y$ an application of Fatou gives $\int 2Y dP \leq \liminf \int (2Y - |X_n - X|) dP$. Linearity of integral and finiteness of the integrals involved gives after cancellation of the Y integral $\limsup \int |X_n - X| dP \leq 0$. This completes the proof.

We shall now show the well-defined-ness of the integral for non-negative simple random variables. Let s be such a random variable. Let $v_1, v_2, \cdots, v_k$ be the distinct values of s and $V_i = s^{-1}(v_i)$ so that we have $s = \sum v_i I_{V_i}$. Consequently $E(s) = \sum v_i P(V_i) = \alpha$, say. We show any representation of s leads to the same value.

Suppose that s is represented as $s = \sum a_i I_{A_i}$ where the sets A_i are non-empty and form a partition of Ω. Clearly then each a_i is a possible value of s and hence must be a v_j for some j. Further $V_j = \sum_{i: A_i \subset V_j} A_i$ and consequently $P(V_j) = \sum_{i: A_i \subset V_j} P(A_i)$. Thus

$$\sum a_i P(A_i) = \sum_j \sum_{i: A_i \subset V_j} a_i P(A_i) = \sum_j v_j \sum_{i: A_i \subset V_j} P(A_i) = \sum_j v_j P(V_j) = \alpha$$

Suppose finally that s is represented as $s = \sum_{i=1}^{m} b_i I_{B_i}$. First we claim that there is a finite partition A_j, $1 \leq j \leq k$ such that each B_i is a union of sets in the partition. In fact, consider all sets of the form $\cap_{i=1}^{m} C_i$ where each C_i is either B_i or its complement. Enumerate the non-empty sets you have so obtained. Clearly you get finitely many such sets , they are disjoint, and each B_i is union of some of these sets and all these sets are in your σ-field. Let these be A_i, $1 \leq j \leq k$. Denote $a_j = \sum_{i:A_j \subset B_i} b_i$. The representation of s tells us that a_j is the value of s on the set A_j. In other words $s = \sum a_j I_{A_j}$ and hence from what was observed earlier, we get $\sum a_j P(A_j) = \alpha$. Note also that $P(B_i) = \sum_{j:A_j \subset B_i} P(A_j)$. As a consequence,

$$\sum_i b_i P(B_i) = \sum_i b_i \sum_{j:A_j \subset B_i} P(A_j) = \sum_j P(A_j) \sum_{i:A_j \subset B_i} b_i = \sum_j a_j P(A_j) = \alpha$$

completing the proof.

Theorem 11 : Take any set $A \in \mathcal{A} = \mathcal{A}_1 \otimes \mathcal{A}_2$

Theorem 12 :

Theorem 13 :

Theorem 14 :

Theorem 15 :

Theorem 16 : We follow Halmos.

(i). For the first part, let (X_n) be independent mean zero random variables with $\sum Var(X_n) < \infty$. For $n \geq 1$, let S_n be the partial sum $\sum_1^n X_i$. Let A be the set of all sample points ω, such that the sequence $(S_n(\omega))$ is not a Cauchy sequence. We just need to show that $P(A) = 0$.

Theorem 17 :

Theorem 18 : For each $n \geq 1$, temporarily, put $\mathcal{F}_n = \sigma\{\mathcal{A}_1, \cdots, \mathcal{A}_n\}$ and $\mathcal{T}_n = \sigma\{\mathcal{A}_{n+1}, \mathcal{A}_{n+2} \cdots\}$.

Theorem 19 :

Theorem 20 (Radon-Nikodym Theorem): The idea is to look at the collection of all random variables X such that $\int_A X d\mu \leq \nu(A)$ holds for all $a \in \mathcal{A}$. There is at least one such, namely, the zero random variable. The hope is that there is a largest such random variable and for that the equality holds for all sets. This is executed as follows.

Let $\alpha = \sup\{\int X d\mu : X \in L\}$ where $L = \{X \geq 0 : \forall A \in \mathcal{A}; \int_A X d\mu \leq \nu(A)\}$. If X and Y are in L then so is $X \vee Y$. This is because if $S = \{X \geq Y\}$ then for any $A \in \mathcal{A}$ we have $\int_{A \cap S} X d\mu \leq \nu(A \cap S)$ and $\int_{A \cap S^c} Y d\mu \leq \nu(A \cap S^c)$ and adding the two inequalities we get $\int_A X \vee Y d\mu \leq \nu(A)$. If $X_n \uparrow X$ and each $X_n \in L$ then $X \in L$. So we can get a $Z \in L$ with $\int Z d\mu = \alpha$. We first claim that if $X \in L$ then $X \leq Z a.e \mu$. If this were not true then just note that $X \vee Z \in L$ and would have expectation larger than α, contradicting the definition of α.

Define a finite non-negative measure on $\mathcal{A}$ by $\lambda(A) = \nu(A) - \int_A Z d\mu$. If only we could show that $\lambda = 0$ then we are done. First we claim that there is an integer $k \geq 1$ and a set A such that $\lambda(A) - \frac{1}{k}\mu(A) > 0$. Indeed if this were not so then for every A and every k, $\lambda(A) - \frac{1}{k}\mu(A) \leq 0$ which implies that $\lambda(A) \leq \frac{1}{k}\mu(A)$ for all k and A showing that λ is actually the zero measure. But then there is nothing to be done, as noted earlier.

accordingly fix a k and A as above, that is, $\lambda(A) > \frac{1}{k}\mu(A)$. In particular $lambda(A) > 0$ and hence $\mu(A) > 0$. How nice if *for every* $B \subset A$, $B \in \mathcal{A}$ we had $\lambda(B) - \frac{1}{k}\mu(B) \geq 0$. Then we could argue that $Z + \frac{1}{k}I_A$ is in L and and larger than Z on the set A of positive μ measure, contradicting an earlier statement about Z. If A does not have this property, we exhibit another set B with this property.

Set $A_0 = A$ and $\beta_0 = \inf\{\lambda(B) - \frac{1}{k}\mu(B) : B \subset A \text{ and } B \in \mathcal{A}\}$.If this is zero we are done. Otherwise take $B_0 \subset A_0$, $B_0 \in \mathcal{A}$ with $\lambda(B_0) - \frac{1}{k}\mu(B_0) < \frac{1}{2}\beta_0$. Set $A_1 = A_0 - B_0$ and do to A_1 what you did for A_0 . Continue. Just note that at each stage you have A_n with $\lambda(A_n) - \frac{1}{k}\mu(A_n) \geq \lambda(A) - \frac{1}{k}\mu(A)$. If you stoped at a finite stage then you got a set B with the property stated in the earlier para. If you continue for ever then the intersection of all the sets A_n would have the desired property.

Uniqueness of such a Z is left to the reader. This completes the proof.

Theorem 21 (Lebesgue decomposition):

Note that $\nu << \mu + \nu$ get the Radon-Nikodym derivative Z. Observe that $Z \leq 1$ w.r.t all these measures. Put $\Omega_1 = [0 \leq Z < 1]$ and $\Omega_2 = [X = 1]$; $\nu_1(A) = \nu(A \cap \Omega_1)$ and $\nu_2 = \nu(A \cap \Omega_2)$ is the required decomposition. If there is another decomposition $\nu = \nu_1^* - \nu_2^*$ then $nu_1 - \nu_1^* = \nu_2 - \nu_2^*$ is both absolutely continuous and singular w.r.t μ (should interpret carefully as these may take negative values) and hence must be zero.

Theorem 22:

Theorem 23 :

Theorem 24 :

Theorem 25 : The plan is the following. We shall get a matrix of sets $(A_{n,\beta})$. These will all be subsets of ω_1. The index n will run over ω where as the index β will run over an uncountable set. Thus there are countably many rows and uncountably many columns. We shall make sure that (i) sets in each row are pairwise disjoint and (ii) sets in a column add upto (that is, union of sets in a column) ω_1 except for a countable set. If this is done, we have a contradiction as follows. By (ii) and the fact that singletons get mass zero, in each column there must be at least one set of positive probability. If you mark those sets in your matrix whose probability is positive, there will be at least one mark in each column. Since there are uncountably many columns, we must have made uncountably many marks. Since there are only countably many rows, there must appear uncountably many marks in some row. By (i), these marked sets are pairwise disjoint and yet all have positive probability $--$ a contradiction.

To get the matrix of sets promised proceed as follows. Say $\alpha \in \omega_1$ is infinite if $\{\beta : \beta \leq \alpha\}$ is infinite, that is, α has infinitely many predecessors. Let I be the set of those α which are infinite. Clearly countably many points of ω_1 are outside I. Indeed by using the property 3 of the set ω_1, consider its first element, say a_0. Denote the first element of $\omega_1 - \{0\}$ by a_1. Having defined $a_0, \cdots, a_n$ put a_{n+1} to be the first element of $\omega_1 - \{a_0, \cdots, a_n\}$. Do this for each natural number n, and put $I' = \{a_n : n = 1, 2 \cdots\}$. Clearly I' is a countable set and any $\alpha \notin I'$ has infinitely many predecessors (I' is a witness) and hence belongs to I.

Now, for each $\alpha \in I$ fix any one-one and onto map $f_\alpha : \{\beta : \beta \leq \alpha\} \mapsto$

$\{0, 1, 2 \cdots\}$. Set

$$A_{n,\beta} = \{\alpha : f_\alpha(\beta) \text{ is defined and equals } n\}$$

If you fix n, these sets are disjoint over β because, f_α being one-one function its value at two different β can not equal n and hence α can not be in two different sets of the row. If you fix β, than for any α such that $\beta \leq \alpha$, f_α is defined at β and equals some n. In other words any such α is in some set of the column, or equivalently, the union of sets in this column misses $\{\alpha : \alpha < \beta\}$ which is countable.

This completes the construction and the proof of the theorem.

PART C : THE PRACTICE

Home Work 1

1. Consider $\Omega = \{0, 1, 2 \cdots\}$. For each $n \geq 0$, let p_n be a nonnegative number such that $\sum p_n = 1$. Define for every subset $A \subset \Omega$, $P(A) = \sum_{n \in A} p_n$. Empty sum is, by definition, zero. Show the following:

   (i) P is a nonnegative function defined for all subsets of Ω.

   (ii) $P(\emptyset) = 0$ and $P(\Omega) = 1$.

   (iii) $P(\cup A_i) = \sum P(A_i)$ for disjoint sequence of sets A_i.

   Conversely, suppose that P is a function defined for all subsets of Ω satisfying the three conditions above. Show that there are indeed nonnegative numbers p_n which add upto unity and the given P is obtained by the formula prescribed above.

   Is it necessary to have Ω to be the set of nonnegative integers in the above problem? What if it were some countable set?

   (the moral is that, when you are dealing with chance experiments that have countably many outcomes, anyone of the following two could be starting points for your modelling : (i) Have the probabilities p_n of outcomes prescribed to you and then *you* define probabilities P of events. (ii) Have probabilities P of all events prescribed to you satisfying the three conditions specified above.)

2. I want to select a number at random from the set Ω (of the above exercise). At random means that chance of selecting number i is same as the chance of selecting the number j, whatever be i and j. Can you build a model for this?

   A model constitutes writing down the set of possible outcomes and associating for each outcome ω a nonnegative number $p(\omega)$ such that all these numbers add to unity. The interpretation is: $p(\omega)$ is the probability of selecting ω. Of course the numbers should reflect that you are selecting at random.

3. Let $\Omega = \{0, 1, 2, \cdots 99\}$. Take a partition Π of this set. Consider the following collection $\mathcal{A}$ of sets: A set $A \in \mathcal{A}$ iff it is a union of sets in the partition. Empty union is, by definition, empty set. Show that this collection of sets is a σ-field of subsets of Ω.

   Conversely, let $\mathcal{A}$ be a σ-field of subsets of Ω. Show that there is indeed a partition Π of Ω such that sets in $\mathcal{A}$ are nothing but unions of sets in the partition.

   Was it important to have exactly 100 elements in Ω? What if Ω were just a finite set? What if it were a countably infinite set? What if it were an uncountable set like the set of real numbers?

4. Suppose that $\mathcal{A}$ is a σ-field of subsets of some set. Assume that there are only finitely many sets which belong to this σ-field. How many sets could there be in the σ- field? Can it have exactly 200 sets? Can it have exactly 1024 sets?

5. Show that the collection of all subsets of a set Ω with the binary operation Δ of symmetric difference forms a group.

6. Let $\{A_n\}$ be a sequence of subsets of a given set Ω. Recall that Lim Inf of this sequence of sets is precisely the set of points that belong to ALL of these sets after some stage, equivalently, those points that fail to belong to finitely many of these sets. Lim Sup is the set of all points that belong to infinitely many of these sets. Recall the definitions of lim inf and lim sup for real numbers a_n. Show that

$$Lim\, Sup\, A_n = \cap_m \cup_{n>m} A_n \quad ; \quad Lim\, Inf\, A_n = \cup_m \cap_{n>m} A_n$$

   What is the analogue for numbers?

   Show that $lim\, inf A_n \subset \lim\, sup A_n$. What is the analogue for numbers?

   State DeMorgans laws for lim sup and lim inf of sets. What is the analogue for numbers?

   You know lim inf and lim sup of functions defined pointwise. What is the relation between Lim sup/ lim inf of sets and Lim sup/lim inf of their indicator functions?

   Suppose that $\mathcal{A}$ is a σ-field of subsets of Ω and all these sets A_n belong to $\mathcal{A}$. Then show that the lim inf and lim sup also belong to $\mathcal{A}$.

7. Let $I_0 = [0,1],\ I_1 = [0, \frac{1}{3}] \bigcup [\frac{2}{3}, 1]$, and so on. In general, if I_n is defined as the union of 2^n disjoint closed intervals each of length 3^{-n}, then I_{n+1} is the union of 2^{n+1} disjoint closed intervals obtained by deleting the middle open one-third interval from each of the 2^n intervals in I_n. Let $C = \bigcap I_n$.

   Show that the sum of the lengths of the deleted intervals adds upto one. However C itself is nonempty. In fact show that C is an uncountable closed set.

   This set C is called a Cantor set. (Do not bother now about my using 'a' instead of 'the' Cantor set, but when we say Cantor set we mean only this set)

8. Define a sequence of functions F_n on R as follows: Each of the functions is 0 on $(-\infty, 0)$ and is 1 on $(1, \infty)$. Here is the definition on $[0,1]$. F_0 is the polygonal line joining the points $(0,0)$ and $(1,1)$. F_1 is the polygonal line joining the points $(0,0), (\frac{1}{3}, \frac{1}{2}), (\frac{2}{3}, \frac{1}{2})$ and $(1,1)$. F_2 is the polygonal line joining $(0,0), (\frac{1}{9}, \frac{1}{4}), (\frac{2}{9}, \frac{1}{4}), (\frac{1}{3}, \frac{1}{2}), (\frac{2}{3}, \frac{1}{2}), (\frac{7}{9}, \frac{3}{4}), (\frac{8}{9}, \frac{3}{4})$ and $(1,1)$. Thus F_n remains flat on each of the intervals deleted so far. You get the picture? Show that the sequence F_n converges uniformly to a continuous function F on R and that F is a probability distribution function. Suppose that μ is the (unique) probability measure corresponding to F. Show tht $\mu(C) = 1$. Show that μ value of singleton sets is zero.

   This distribution function F is called the Cantor distribution function and the probability μ is called the Cantor measure.

9. Let us make a notation to save repeating long phrases. The collection of open subsets of the real line is denoted $\mathcal{G}$ and the collection of closed sets by $\mathcal{F}$. For *any* class of sets $\mathcal{C}$, let us denote by $\mathcal{C}_\sigma$ the collection of set

obtained by taking countable union of sets in the class $\mathcal{C}$ and let us denote by $\mathcal{C}_\delta$ the collection of sets obtained by taking countable intersections of sets in the class $\mathcal{C}$.

For example $(0,1)$ is in $\mathcal{G}$ but not in $\mathcal{F}$ where as $[0,1]$ is in $\mathcal{F}$ but not in $\mathcal{G}$.

Since union of open sets is again open, observe $\mathcal{G}_\sigma$ is just the class $\mathcal{G}$ itself. Similarly the class $\mathcal{F}_\delta$ is the class $\mathcal{F}$ itself.

The set Ir of irrational numbers is in $\mathcal{G}_\delta$ but not in $\mathcal{F}_\sigma$ where as the set Q of rational numbers is in $\mathcal{F}_\sigma$ but not in $\mathcal{G}_\delta$.

If you take the set A to be the union of the two sets : irrationals in $[0,1]$,rationals in $[10,11]$. This set is in none of the above classes. Indeed this set is in $\mathcal{G}_{\delta\,\sigma}$.

Home Work 2

10. Suppose that P is a probability on $(R,\mathcal{B})$ with distribution function F.

    For any point a, show that $P\{a\} = F(a) - F(a-)$.

    Show that F is continuous at a point a iff $P\{a\} = 0$.

    Show that F is continuous iff F is uniformly continuous.

    Show that the set $D = (x : P\{x\} > 0)$ is a countable set.

    A probability P is said to be *continuous* if F is continuous, equivalently, for every $x \in R$, $P\{x\} = 0$. A probability P is said to be *discrete* iff there is a countable set D such that $P(D) = 1$.

    Give examples of probabilities which are continuous, Give examples which are discrete, Give examples of probabilities which are neither continuous nor discrete. Are there examples of probabilities which are both continuous and discrete?

    If P_1 and P_2 are two probabilities on $(R,\mathcal{B})$ and if $0 \le c \le 1$ we can define their convex combination as follows: for every Borel set B, $P(B) = c\,P_1(B) + (1-c)\,P_2(B)$. Show that this is again a probability on $(R,\mathcal{B})$.

    Given any probability P on R show that there is exactly one discrete probability P_1 and exactly one continuous probability P_2 and exactly one number c in $[0,1]$ such that $P = cP_1 + (1-c)P_2$.

11. Suppose that P is a probability on $(R,\mathcal{B})$. Show that given a Borel set B and a number $\epsilon > 0$, we can get a closed set C and an open set U such that $C \subset B \subset U$ and $P(U - C) < \epsilon$.

    In particular, if you want to calculate probability of the Borel set B, then you can take the supremum of P values of closed sets C contained in B. One can do better. Show that for any Borel set B, $P(B) = Sup\,\{P(K) : K \text{ compact } \subset B\}$. So to calculate $P(B)$ you can calculate the supremum of P values of compact subsets of B.

12. Let $\Omega = R$ and $\mathcal{F}$ be the collection of all sets $A \subset \Omega$ such that either A is finite or $\Omega - A$ is finite. Show that this is a field of subsets of Ω. Is it a σ-field? What if Ω were the set of integers instead of R? What if Ω were a finite set?

    Consider the above field. Define $P(A) = 0$ or 1 according as A is a finite set or not. Show that P is finitely additive probability but NOT countably additive.

13. Let $\Omega = R$ and $\mathcal{A}$ be the collection of sets $A \subset R$ such that either A is countable or $\Omega - A$ is countable. Show that this is a σ-field of subsets of Ω. Define $P(A) = 0$ or 1 according A is countable or not. Show that this is indeed a probability on $\mathcal{A}$.

14. It is interesting to note that the probability in the above example takes only two values zero and one, but yet the probability of each singleton set is zero. It is natural to ask if I should cook up such apparently useless σ-fields to exhibit $0-1$ valued probabilities. The answer is: yes!

    More precisely show the following : Let P be a probability on the Borel σ-field $\mathcal{B}$ of R. Suppose that P takes only two values 0 and 1. Then show that there is a number $a \in R$ such that $P\{a\} = 1$. In particular for any Borel set B, $P(B)$ is 1 or 0 according as the Borel set includes a or not.

    What is so special about R ? Nothing. We shall discover this slowly. First some practice with the Borel σ-field.

15. Consider $\Omega = R$. Let $\mathcal{C}$ be any one of the following collections of subsets of R : all intervals $(-\infty, a)$ OR all intervals (a, ∞) OR all intervals $(-\infty, a]$ OR all intervals $[a, \infty)$ OR all intervals (a, b) OR all intervals $(a, b]$ OR all intervals $[a, b)$ OR all intervals $[a, b]$ OR the same classes with a and/or b restricted to rationals OR the collection of all open sets OR the collection of all closed sets. Show that the the Borel σ-field $\mathcal{B} = \sigma(\mathcal{C})$. What if I replaced the word rationals by D where D is a fixed countable dense subset of R.

16. You must be wondering that I am obsessed with R and σ-fields on R. Not really. Consider the cartesian plane R^2. Consider the class of sets $A \times B$ where A and B are Borel subsets of R. Denote this class by $\mathcal{R}$. These are called Borel rectangles. Denote $\sigma(\mathcal{R})$ by $\mathcal{B}^2$ OR sometimes by just $\mathcal{B}$. This is called the Borel σ-field on R^2.

    Show that the open unit disc belongs to $\mathcal{B}^2$. Show that every open subset of R^2 is in $\mathcal{B}^2$. Show that every closed set also belongs. In particular, the sets $\{(x, y) : x \leq y\}$; $\{(x, y) : x \geq y\}$; $\{(x, y) : x = y\}$; $\{(x, y) : 5x+21y = 93\}$ are all Borel subsets of R^2.

    Fix any one of the classes $\mathcal{C}$ of the previous exercise. Let $\mathcal{R}^1$ be the collection of sets $A \times B$ where $A, B \in \mathcal{C}$. Just to get practice show that $\mathcal{B}^2 = \sigma(\mathcal{R}^1)$

    What if I fixed two of the classes $\mathcal{C}$ and $\mathcal{D}$ of the previous exercise and considered $\mathcal{R}^1$ to be the collection of sets $A \times B$ where $A \in \mathcal{C}$ and $B \in \mathcal{D}$. Will $\mathcal{B}^2 = \sigma(\mathcal{R}^1)$?

17. Say that a σ-field $\mathcal{A}$ is *countably generated* if there is a countable family $\mathcal{C}$ of sets such that $\mathcal{A} = \sigma(\mathcal{C})$.

    For instance the Borel σ-field on R is countably generated. The power set on integers is a countably generated σ-field. Do you think the countable - cocountable σ-field described earlier is countably generated? (No,not a tongue twister) The Borel σ-field on R^2 is also countably generated.

    Countably generated σ-fields are not difficult to obtain. You can start with your own choice of set Ω and with your own favourite countable collection $\mathcal{C}$ of subsets of Ω and just consider $\sigma(\mathcal{C})$.

    Suppose that $\mathcal{A}$ is a countably generated σ-field of subsets of Ω, generated by , say, $\mathcal{C} = \{C_1, C_2, \cdots\}$. For each i take C_i OR complement of C_i – you make your choice. Denote it by B_i. Let $A = \cap_i B_i$. Clearly $A \in \mathcal{A}$. Suppose that $A \neq \emptyset$. Then A is one of the smallest sets available in the σ-field. More precisely show that if $A_0 \in \mathcal{A}$ is a proper subset of A then A_0 must be the empty set. Such sets are called *atoms* of the σ-field.

    To continue the story, suppose now that P is a probability on $\mathcal{A}$ and P is $0-1$ valued. Show that there is an atom A such that $P(A) = 1$. In particular if $\mathcal{A}$ is countably generated and has singleton sets, then any $0-1$ valued probability P must be concentrated at a point – that is, there is a point a such that $P(A) = 1$ or 0 according as A contains the point or not.

Home Work 3

18. Suppose that F is the uniform $(0\,,\,1)$ distribution function and P is the probability associated with it on the Borel σ-field of the real line. Show that for every subinterval $(a, b) \subset (0, 1]$ we have

$$P(a, b) \;=\; P(a, b] \;=\; P[a, b) \;=\; P[a, b] \;=\; b - a$$

    In otherwords P extends the concept of length to all Borel subsets of the unit interval. If you like you can still call it length. What is the length of the set of rationals? What is the length of the set of irrationals? What is the length of a point? What is the length of the Cantor set? Of course this P 'sits' only on the unit interval. This means complement of the unit interval gets P value zero. Show this.

    It is sensible to ask if we can extend the concept of length on the real line itself, instead of only on the unit interval and moreover if such an extension exists whether it is useful at all. We have to be careful. The length of the interval $[5, 10]$ is 5. So the extension, if exists, will not be probability. The answer is Yes, such extension exists and is indeed useful. Be prepared to see something which is NOT a probability.

19. If A is a Borel set and x is any real number, show that translate of A, defined by $A + x = \{y + x : y \in A\}$ is also a Borel set. Show that $-A = \{-y : y \in A\}$ is also a Borel set.

20. Denote the Probability we defined earlier by P_0 instead of P, just for this exercise. For any integer n positive or negative define $P_n(A) = P_0(A-n)$. Show that this is also a probability on the Borel σ-field $\mathcal{B}$ of the real line. It sits on the interval $(n\,,\,n+1]$. Show that its distribution function is nothing but the uniform $(n\,,\,n+1]$ distribution function.

    Define now for any Borel set A, $\lambda(A) = \sum_n P_n(A)$. Please understand that it is a sum over all integers. More specificaly, you know sums like $\sum_{n=1}^{\infty} x_n$ and I am here talking about $\sum_{n=-\infty}^{\infty} x_n$. So make yourself clear what exactly we mean before proceeding.

    Show that λ so defined on $\mathcal{B}$ satisfies the following conditions: $\lambda(\emptyset) = 0$ and $\lambda(\cup_i A_i) = \sum \lambda(A_i)$ for disjoint Borel sets A_i. In otherwords λ satisfies the axioms of probability **except** that value for whole space is 1. Such objects go by the name of *measure* as opposed to *probability*

    This λ is called Lebesgue measure on the real line. Show that this extends the concept of length. This means show that for any $a < b$

$$\lambda(a,b) \;=\; \lambda(a,b] \;=\; \lambda[a,b) \;=\; \lambda[a,b] = b-a$$

    Show that λ is translation invariant, this means for any A and any real number x, $\lambda(A) \;=\; \lambda(A+x)$.

21. Suppose Ω is a set and $\mathcal{A}$ is a σ-field of subsets of Ω. A measure μ on $\mathcal{A}$ is a function that associates to every set $A \in \mathcal{A}$ a number $\mu(A)$ such that

$$0 \le \mu(A) \le \infty, \qquad \mu(\emptyset) = 0$$

$$\mu(\cup A_i) \;=\; \sum \mu(A_i) \quad \text{for disjoint sets } A_i$$

    Since we are allowing μ to take the value ∞ and we are talking about sums we should agree to an arithmetic involving $+\infty$. It is simply this: Any thing added to ∞ is ∞. (Just remember we are not talking about $-\infty$). We agree never to subtract ∞ from ∞.

    Of course the definition is too general to be of much use. For example show that μ defined by $\mu(A) = 0$ or ∞ according as A is empty or not defines a measure. This is not too interesting. Of course you may say that μ defined by $\mu(A) = 0$ for all sets A is not interesting too. For some good reasons this is not that uninteresting.

    A measure is said to be *finite* if $\mu(\Omega) < \infty$. A measure μ is said to be *σ-finite* if Ω is a union of countably many sets Ω_i from the σ-field each having finite μ value .

    Finite measures are easily related to probabilities. If P is a probability on $(\Omega, \mathcal{A})$, and c is any nonnegative real number then μ defined by $\mu(A) = c\,P(A)$ for sets $A \in \mathcal{A}$ is a finite measure. Conversely, suppose μ is a finite measure on $(\Omega, \mathcal{A})$, then either μ is the identically zero measure OR there is indeed a nonnegative number c and a probability P such that $\mu = c\,P$ as above. In other words finite measures are nothing but 'multiples ' of probabilities.

22. Show that the Lebesgue measure is a σ finite measure. Show that if μ is a translation invariant measure on the Borel sets of real line with $\mu[0,1] = 1$, then it must be the λ above. If I do not assume $\mu[0,1] = 1$ then there are two possibilities : either $\mu[0,1] < \infty$ OR $\mu[0,1] = \infty$. In the first case show that μ is indeed a constant multiple of our good old λ. In the second case there are several subpossibilities. For example the μ that gives values 0 or ∞ according as the set is empty or not is one such. the μ that gives values 0 or ∞ according as the set is countable or not is one such . We leave the story here.

23. Suppose that μ is a finite measure on the Borel σ-field of the reals. Define its distribution function by $F(x) = \mu(-\infty, x]$. Show that this is right continuous, nondecreasing, has limit zero at $-\infty$ and a finite limit at $+\infty$. Show conversely that given any function with these properties, there is a unique finite measure μ on the Borel sets such that $\mu(-\infty, x] = F(x)$ for each real number x.

    Suppose, you try to do the same thing with non finite measures, is there a hope of succeeding? More specifically suppose you looked at the distribution function of the Lebesgue measure what do you get? Convince yourself that we should change our method of attack.

24. Suppose that μ is a measure on the Borel σ-field of the real line which gives finite value for finite intervals. That is, for any reals $a < b$, we have $\mu(a,b) < \infty$. Incidentally such measures are called *Radon measures*. Define its *rdf* (Radon distribution function as opposed to probability distribution function) by $F(x) = \mu(0,x]$ for $x > 0$ and $-\mu(x,0]$ for $x \leq 0$. This function is no longer nonnegative, perhaps. However, show that this is nondecreasing; right continuous; $F(x) \leq 0$ for $x \leq 0$; $F(x) \geq 0$ for $x \geq 0$. (consequently, of course, $F(0) = 0$.) It is also true that any such function gives a unique Radon measure. You need not workout the details but think how to get it.

    Are you wondering what all these boring (?) exercises are telling us? Remember it is easy to cook up functions on the real line rather than prescribing probabilities / measures on all Borel sets.

    There are several other ways of constructing probabilities / measures. We shall learn them slowly.

Home Work 4

25. Let $\mathcal{A} = \{\emptyset, \Omega\}$. Show that a function on Ω is a random variable if and only if it is a constant function.

26. Let $A \subset \Omega$ and $\mathcal{A} = \{\emptyset, A, A^c, \Omega\}$. Show that a function is a random variable iff it is a constant on A and it is also a constant on A^c. Generalize to finite σ-fields.

27. Let $(\Omega, \mathcal{A})$ be given. Let ω_1 and ω_2 be two points in Ω. Suppose that they move together in $\mathcal{A}$ – this means that for every set $A \in \mathcal{A}$, either both ω_1 and ω_2 are in A OR both are outside A. Show that every random

variable X has the property $X(\omega_1) = X(\omega_2)$. In particular show that every random variable must be constant on any atom of the σ-field. (Of course, atoms may not exist. but that is a different issue).

Conversely show that if every random variable X has the property $X(\omega_1) = X(\omega_2)$, then ω_1 and ω_2 move together in $\mathcal{A}$.

28. Let $\mathcal{A}$ be the countable -cocountable σ-field. Show that a function is random variable if and only if there is a countable set *outside* of which it is a constant.

29. Let Ω be the real line and $\mathcal{A}$ be the Borel σ-field. Show that every continuous function is a random variable. Show that every right continuous function is random variable. Show that every left continuous function is a random variable. Show that every monotone function is a random variable. Suppose that f is *any* real valued function. Show that the set of those points x where f is continuous is a Borel set.

30. We discussed in the class that we must allow functions that may take values $\pm\infty$. Let $(\Omega, \mathcal{A})$ be given.

    Extended real line means the set R of real numbers along with two animals denoted $+\infty$ and $-\infty$. The understanding is that for any $x \in R$, $-\infty < x < +\infty$ and $-\infty < +\infty$. We denote $+\infty$ as ∞ also omitting the symbol $+$. The extended real line is denoted by $\overline{R}$.

    Let X be a function on Ω to the extended real line. Say that X is a random variable if for each real number a the set $(X < a)$ is in $\mathcal{A}$. To avoid confusion, these are sometimes called *extended random variables*. But we shall not use this term.

    If X is a *rv* then show that the sets $[X = -\infty]$, $[X = +\infty]$ $[-\infty < X < +\infty]$ are all in $\mathcal{A}$.

    Show that the max, min of two *rv* is again a *rv*. Show that the infimum, supremum of a sequence of *rv* is again a *rv*. Show that the *liminf* and *limsup* of a sequence of *rv* is again a *rv*.

    For the lattice operations above you need not know the arithmetic of $\overline{R}$. Here is the arithmetic: Let $x \in \overline{R}$. (i) if $x \neq -\infty$ then $x + \infty = \infty$. (ii) If $x \neq +\infty$ then $x - \infty = -\infty$. (iii) $x \times 0 = 0$. (iv) for $x > 0$, $x \times \pm\infty = \pm\infty$ whereas for $x < 0$, $x \times \pm\infty = \mp\infty$. We shall NOT talk about $\infty - \infty$.

    Show that if X is *rv* then so is $-X$. Show that if X and Y are *rv* and if at NO point they take infinities of opposite signs as values, then $X + Y$ is defined and is a *rv*. Show that $X.Y$ is also a *rv*.

    Let us still use the good old definition of simple *rv*. In other words a simple *rv* does not take infinite values. Show that for any positive *rv* X there is a sequence of simple *rv* $\{X_n\}$ which pointwise increase to X. Show that for any *rv* X there is a sequence of simple *rv* $\{X_n\}$ converging pointwise to X.

31. Let X and Y be two *rv*. Show that the sets $[X = Y]$, $[X < Y]$, and $[X > Y]$ are all in $\mathcal{A}$. If the random variables are just real valued then you could just look at inverse image under $X - Y$ of $\{0\}$, $(-\infty, 0)$ etc.

The main point is that even if the variables take infinite values, you can still prove the result.

I think we talked enough for now about *rv*. Before it gets boring let us quickly change to something else.

32. If you want some sets necessarily in your σ-field then you can consider the σ-field generated by your sets. What if you want a specific function to be a random variable? Can you achieve by taking a suitable σ-field? The answer is: yes.

    Let Ω be any set and f be *any* real valued function on Ω. Consider the collection $\mathcal{A}$ of set $f^{-1}(B)$ where B ranges over all Borel subsets of the real line. Show that this collection is a σ-field of subsets of Ω. Show that f is a random variable w.r.t this σ-field. Show further that if there is any σ-field $\mathcal{D}$ on Ω w.r.t which f is a random variable, then $\mathcal{A} \subset \mathcal{D}$. In other words $\mathcal{A}$ is the smallest σ-field that makes f a random variable. This is called the σ-field generated by f, denoted by $\sigma(f)$.

    What if I want two functions f and g to be random variables?. Consider the collection $\mathcal{C}$ of all sets of the form $f^{-1}(B)$ as well as sets of the form $g^{-1}(B)$ where B is a Borel subset of R. Of course this collection need not be a σ-field. Consider the σ-field generated by this class. Show that this is the smallest σ-field of subsets of Ω w.r.t which both f and g are *rv*. This is called the σ-field generated by $\{f, g\}$, denoted by $\sigma(f, g)$.

    What if we have a class $\mathbf{F}$ of functions instead of one or two? Define $\sigma(\mathbf{F})$.

    Suppose that I have a class $\mathcal{C}$ of subsets of Ω. Consider the class $\mathbf{F}$ consisting of indicator functions of sets in $\mathcal{C}$. There is no difference between a set and its indicator function – in the sense that if you know the set you can calculate the value of indicator function at any point; conversely if you can calculate the values of the indicator function at every point then you know what is the set. It is reasonable to conjecture that $\sigma(\mathcal{C}) = \sigma(\mathbf{F})$. Prove this.

33. I have a set Ω and $\mathcal{A}$ a σ-field of subsets of Ω. Unfortunately, a set $B \subset \Omega$ which I like is not in $\mathcal{A}$. Of course I should look at the smallest σ-field generated by the collection of sets in $\mathcal{A}$ and also the set B. Denote this σ-field by $\overline{\mathcal{A}}$.

    The question is whether we can explicitly describe sets in this new $\overline{\mathcal{A}}$. In the description we can make use of sets in $\mathcal{A}$. Yes, here is how. Take two sets A_1 and A_2 in $\mathcal{A}$. Show that the set $(A_1 \cap B) \cup (A_2 \cap B^c)$ is in $\overline{\mathcal{A}}$. Show that every set in $\overline{\mathcal{A}}$ is of this form.

34. You have seen finite σ-fields. You have seen σ-fields with uncountably many sets. So far you have not seen any σ-field that has only countably infinitely many sets in it. (Understand. I am NOT saying that each set in the σ-field should be a countable set. I want the number of sets in the σ-field to be countably infinite). Show that you can not find any such. More precisely, show that if a σ-field has an infinite number of sets in it then it must have an uncountable number of sets in it.

Home Work 5

In what follows $(\Omega, \mathcal{A}, P)$ is a probability space. First order of business is to convince ourselves that our present definition of expectation is same as what we used in BStat. Remember that in BStat you did not bother as to where the random variable is defined. Secondly the expectation is defined for discrete random variables as appropriate sum and for random variables having density as an appropriate integral.

35. Suppose that X is a random variable that takes finitely many values say x_i , $1 \leq i \leq k$. Suppose that $P[\omega : X(\omega) = x_i] = p_i$. So your B-def says $EX = \sum x_i\, p_i$. Prove this (with our definition)

    Suppose that X takes countably many values x_i and $P[\omega : X(\omega) = x_i] = p_i$. B-def says that EX exists iff $\sum |x_i|\, p_i < \infty$ and in such a case $EX = \sum x_i p_i$. Prove this (with our def).

    A little tricky point : I do not tell you that X takes countably many values. But I do tell you that there are countably many values x_i such that $\sum p_i = 1$ where $P[\omega : X(\omega) = x_i] = p_i$. Show that EX according to our present def exists iff $\sum |x_i|\, p_i < \infty$ and in that case $EX = \sum x_i\, p_i$.

36. Suppose that X is a random variable such that it has a density $f(x)$. What does this mean? It means if you take any real number a then $P[\omega : X(\omega) \leq a] = \int_{-\infty}^{a} f(x)dx$. Here the integral is the usual Riemann integral that you learned in BStat Calculus course and used it in probability to calculate expectations.

    First assume that f is bounded continuous on an interval and zero outside that interval (like the uniform density). Show that EX as defined by us indeed equals $\int x\, f(x)\, dx$.

    Consider tha case when f may not be continuous, but bounded and piecewise continuous on a bounded interval and zero outside (like for example f equals $1/2$ on unit interval and $1/8$ on the interval 9 to 13 and zero elsewhere). Show that EX as defined by us indeed equals $\int x\, f(x)\, dx$.

    Consider the case when f is exponential density with parameter one. Show that EX as defined by us indeed equals $\int x\, f(x)\, dx$.

    Consider the case when X has standard normal density. Show that EX as defined by us indeed equals $\int x\, f(x)\, dx$.

    Suppose that X has standard Cauchy density. Show that X is not integrable.

    Well if you got the idea, it is good enough to convince you that B-def is same as our def. No need to complicate life by proving general theorem at this stage.

37. Suppose that there is a point $\omega^* \in \Omega$ such that P is concentrated at ω^* – This means $P(A)$ is one or zero according as A includes the point ω^* or not. Show that an extended *rv* X is integrable iff $X(\omega^*) \neq \pm\infty$ and in such a case $E(X) = X(\omega^*)$.

38. Suppose that there are finitely many distinct points $\omega_1, \omega_2, \cdots \omega_k$ and numbers $p_1, p_2, \cdots, p_k$ strictly positive adding to one such that $P(A) = \sum_{\omega_i \in A} p_i$ for $A \in \mathcal{A}$. Show that an extended random variable X is integrable iff for each i, $X(\omega_i)$ is finite and in that case $E(X) = \sum X(\omega_i)p_i$.

    What if P is concentrated at countably many points?

39. Let us now take the specific example of $\Omega = R$, $\mathcal{A}$ to be the Borel sigma field, P to be the uniform (0,1) probability. Show that if X is a random variable and $X(\omega) = 0$ for $0 < \omega < 1$, then X is integrable and $E(X) = 0$. Thus values on the interval (0,1) only matter!

    Suppose that X is continuous and bounded on the interval (0,1). Show that X is integrable and its integral is same as the Riemann integral $\int_0^1 X(\omega)\, d\omega$. If you feel uncomfortable with the notation, change our agreement; temporarily denote points by x instead of ω and rv by f instead of X so that the last integral appears as familiar object $\int_0^1 f(x)\, dx$. *But* whatever notation you use its meaning and value remain the same!

    What if I do not assume continuity of X, but only say that it is bounded measurable and Riemann integrable on (0,1).

    There are more random variables which you can integrate now and you could not do earlier. Here is one : Take the indicator function of the set of rationals. Show that it is not Riemann integrable, but our definition makes it integrable. What is its integral?

40. Given any probability P on $(R, \mathcal{B})$ show that there is a closed set S with the following two properties: (i) $P(S) = 1$ (ii) if C is a proper closed subset of S then $P(S) < 1$. Thus S is the smallest closed set with full probability. Can there be two such sets? No. Prove it. This S is called the support of P.

    What is the support of uniform (0,1) probability? What is the support of standard normal probability? What is the support of binomial or Poisson probability?

    If I take a probability that gives strictly positive mass to each rational number then what will be its support?

41. Suppose that μ is a σ-finite measure on $(\Omega, \mathcal{A})$. Go through the four clauses of integration. Remember simple functions are *still* real valued. Develop properties of integral for extended random variables.

    This will show if you understood the definitions in the class or not.

42. Here is a nice space – favourite for probabilists.

    Let 2^ω consist of the space of sequences of zeros and ones – called coin tossing space (why?). For any finite sequence $s = (s_1, s_2, \cdots s_k)$ of zeros and ones let A_s be the set of all those sequences in our space for which the first k coordinates are as prescribed. Here $k \geq 0$ is any integer. This is an event whose occurrence can be decided as soon as you have tossed the coin k times. You need not wait until all the tosses are made.

    Let $\mathcal{A}$ be the σ-field on 2^ω generated by all these sets. Show that there is a unique probability P on this σ-field such that for any s of length k,

$P(A_s) = 2^{-k}$. This is called the fair coin tossing probability. What about biased coin tossing probability?

Home Work 6

43. Let $(\Omega, \mathcal{A}, \mu)$ be a σ-finite measure space. Suppose that Z is an integrable random variable. Define for every set $A \in \mathcal{A}$, $\nu(A) = \int Z.I_A \, d\mu$. Of course ν is not necessarily positive.

    Show that $\nu(\emptyset) = 0$ and ν is countably additive – that is, for any sequence of disjoint $A_n \in \mathcal{A}$ we have $\nu(\cup A_n) = \sum \nu(A_n)$. Such things are called signed measures. Let $\Omega^+ = [Z \geq 0]$ and $\Omega^- = [Z < 0]$. Then both these sets are in $\mathcal{A}$. Put for any $A \in \mathcal{A}$, $\nu^+(A) = \nu(A \cap \Omega^+)$ and $\nu^-(A) = \nu(A \cap \Omega^-)$.

    Show that both these are positive finite measures and $\nu = \nu^+ - \nu^-$; just like decomposition of X into X^+ and X^-. Indeed ν^+ is called the positive part of ν and ν^- is called the negative part of ν.

    The set Ω^+ is called the positive set for ν and Ω^- is called the negative set for ν. Sometime back I told you that there is use and merit in considering finite as well as σ-finite measures though our interest is in only probabilities. Similarly, there is use as well as merit in considering signed measures – that is, measures that may take positive and negative values. Of course we shall not indulge in such things.

44. In the above exercise if we have in particular $Z \geq 0$ and $\int Z \, d\mu = 1$ then of course your ν will be a probability. This is one very powerful way of getting probabilities. In this case Z is called the density of ν w.r.t μ.

    Take the special case Ω to be the real line , $\mathcal{A}$ to be the Borel σ-field and μ to be the Lebesgue measure.

    Take $Z = \frac{1}{\sqrt{2\pi}} exp(-x^2/2)$. Show that $\int Z \, d\mu = 1$. Which probability are we talking about?

    Take $Z = x^{a-1}(1-x)^{b-1} \, I_{(0,1)}$. Show that $\int Z \, d\mu$ is finite. Thus by taking a multiple of Z we do get a probability. Which probability are we talking about? here $a > 0$ and $b > 0$.

    Take the special case Ω to be the set of non-negative integers, $\mathcal{A}$ to be the collection of all subsets and μ to be the counting measure – that is, $\mu(A)$ is number of points in A. Observe that this is σ-finite. Take $Z(k) = e^{-5} \, 5^k/k!$. Show that $\int Z \, d\mu = 1$. Which probability are we talking about?

    In the same setup take $Z(k) = \binom{n}{k} p^k \, (1-p)^{n-k}$ for $k \leq n$ and zero for $k > n$. Show that $\int Z \, d\mu = 1$. What are we talking about? Here $0 < p < 1$ and $n \geq 1$ is an integer.

    As a final special case, suppose I take Ω to be the set of nonnegative integers and $\mathcal{A}$ to be the collection of all subsets – just as above. Instead of counting measure, I take $\mu(A) = \sum_{k \in A} \, e^{-5} \, 5^k/k!$. This is a probability. Take $Z(k) = \binom{20}{k} (0.3)^k \, (0.7)^{20-k} \, e^5 \, k! \, 5^{-k}$ for $k \leq 20$ and zero for $k > 20$. Show $\int \, Z \, d\mu = 1$. Which probability are we talking about?

Thus Binomial probability has a density w.r.t Poisson probability. Do you think that the Poisson probability has a density w.r.t the Binomial probability?

The idea is not to complicate well known discrete distributions, but to impress upon you that there is no distinction between discrete and continuous distributions. All are *probabilities*. One more point to note is that in BStat when you said density f it was implicit that you are talking density w.r.t to Lebesgue measure. It was not mentioned explicitly. As you see the Poisson distribution has density too – but not w.r.t Lebesgue measure, but w.r.t the counting measure on integers. What does the last special case say?

45. Let $(\Omega, \mathcal{A}, \mu)$ be a σ-finite measure space. Suppose that Ω' is a non-empty set with a σ-field of subsets $\mathcal{A}'$. Suppose that T is a function defined on Ω into Ω' such that for each set $B \in \mathcal{A}'$ we have $T^{-1}(B) \in \mathcal{A}$. Such a map is called measurable map from $(\Omega, \mathcal{A})$ to $(\Omega', \mathcal{A}')$. Define ν on $\mathcal{A}'$ by $\nu(B) = \mu(T^{-1}(B))$.

    Show that ν is a measure on $\mathcal{A}'$ – that is $\nu(\emptyset) = 0$ and that ν is countably additive.

    If X is an extended random variable on $(\Omega', \mathcal{A}')$, show that the composed map $X(T(\omega))$, denoted by XoT is an extended random variable on Ω.

    Show by example that ν need not be σ-finite. *From now on assume that ν is σ-finite.*

    Show that for any random variable X on Ω' , $\int X\, d\nu$ is defined if and only if $\int XoT\, d\mu$ is defined and in such a case both are equal.

    If μ were a probability then ν is also probability. This is another powerful method of cooking probabilities. The above formula of calculating integrals w.r.t ν in terms of integrals w.r.t μ is called the change of variable formula.

46. A special case of the previous exercise arises when μ is a probability and Ω is same as Ω'. Then ν is also defined on the same space as μ. A further special case arises if we moreover have a situation when this new probability ν has density w.r.t μ. The Jacobian formula you learnt is nothing but an explicit computation of the density in such a situation. To understand matters the best is an example.

    Let Ω be the unit interval $[0, 1]$ with its Borel σ-field $\mathcal{B}$ and μ be the uniform probability – which is, of course, the Lebesgue measure. As a consequence whenever I write $d\mu(x)$ you can as well think of usual dx. Carefully understand what we have here. I shall change our notation temporarily to make you feel comfortable with BStat notation. Take $T(x) = x^2$. The induced probability be ν. The change of variable formula tells you that for any bounded random variable f on our space

$$\int f\, d\nu = \int f(x^2)\, d\mu.$$

    Taking $Z = 1/(2\sqrt{y})$ show that for any interval $A = (a, b)$ we have $\nu(A) = \int ZI_A\, d\mu$. Hence deduce that for any Borel set A , $\nu(A) = \int ZI_A\, d\mu$. In

otherwords the probability ν has density w.r.t μ and the density is the Z above. So remember, from an earlier exercise, $\int f\,d\nu = \int f.Z\,d\mu$. Use this in our earlier equation to get

$$\int f.Z\,d\mu = \int f(x^2)\,d\mu.$$

Deciphering the notation you get

$$\int f(x^2)\,dx = \int f(y)\frac{1}{2\sqrt{y}}\,dy.$$

If you substitute $x^2 = y$ on left side, so that the jacobian is $dx = 1/(2\sqrt{y})dy$, you get the above formula by Jacobian rule you learnt. Now think what we did.

47. Let $(\Omega, \mathcal{A}, \mu)$ be a σ-finite measure space.

    Suppose that X is a random variable on Ω such that $\mu(X \neq 0) = 0$. Show that $\int X\,d\mu$ exists and equals zero.

    Suppose that X is a nonnegative random variable such that $\mu(X > 0) > 0$. Show that $\int X\,d\mu > 0$.

    Suppose that X and Y are two integrable random variables such that for every set $A \in \mathcal{A}$, we have $\int_A X\,d\mu = \int_A Y\,d\mu$. Then show that $\mu(X \neq Y) = 0$. Here $\int_A X\,d\mu$ means $\int X\,I_A\,d\mu$.

48. Let $(\Omega, \mathcal{A}, \mu)$ be a σ-finite measure space. Define $\mathcal{N} = \{N \in \mathcal{A} : \mu(N) = 0\}$ and $\mathcal{I} = \{B : B \subset N \text{ for some } N \in \mathcal{N}\}$.

    Show that if $I \in \mathcal{I}$ and $J \subset I$, then $J \in \mathcal{I}$.

    Show that if $I_n \in \mathcal{I}$ for each n, then $\cup_n I_n \in \mathcal{I}$.

    Define $\mathcal{A}_0 = \{A\Delta I : A \in \mathcal{A}\,,\ I \in \mathcal{I}\}$. Show that this is a σ-field of subsets of Ω and it includes $\mathcal{A}$.

    If a set in $\mathcal{A}_0$ can be written in two ways as $A_1\Delta I_1$ and $A_2\Delta I_2$ with each A_i in $\mathcal{A}$ and each I_i in $\mathcal{I}$, then show that $\mu(A_1\Delta A_2) = 0$; in particular $\mu(A_1) = \mu(A_2)$.

    For $A_0 = A\Delta I \in \mathcal{A}_0$ where as usual $A \in \mathcal{A}$ and $I \in \mathcal{I}$, define $\tilde{\mu}(A_0) = \mu(A)$. Show that this is well defined and is a σ-finite measure on $\mathcal{A}_0$ and this extends μ.

    This space $(\Omega, \mathcal{A}_0, \tilde{\mu})$ has the following interesting property: If $A \in \mathcal{A}_0$, $\tilde{\mu}(A) = 0$ and $B \subset A$ then $B \in \mathcal{A}_0$ and $\tilde{\mu}(B) = 0$.

    Such a measure space is called complete measure space. If you have length/probability of an event to be zero then it is only natural to feel that, first of all, every subset of it must have length/probability and moreover the length/probability should be zero. The only problem is that subsets of such events may not be in our σ-field. This exercise tells you that your natural feeling is vindicated by taking a bigger model without spoiling anything that exits so far.

49. Consider Ω to be the real line and $\mathcal{A}$ to be the Borel σ-field. Suppose that $\mathbf{F}$ is a collection of random variables such that (i) every continuous function is in $\mathbf{F}$ and (ii) if each X_n is in the family $\mathbf{F}$ and if X_n converges pointwise to X then $X \in \mathbf{F}$.

    (a) Show that if $X \in \mathbf{F}$ then $-X \in \mathbf{F}$. (b) Show that if X and Y are in $\mathbf{F}$ then their sum is also in $\mathbf{F}$. (c) Show that $I_{[0,1]} \in \mathbf{F}$. (d) Show that every random variable is in $\mathbf{F}$.

    Replace (i) above by (i') : Every C^∞ function with compact support is in $\mathbf{F}$. Prove all the above four statements.

    Replace (i) by (i') and (ii) by (ii'): if each X_n is in the family $\mathbf{F}$ and this sequence is uniformly bounded and if X_n converges pointwise to X then $X \in \mathbf{F}$. Show that (a), (b), (c) are still correct. Show that (d) may not be correct. Show however that (d') is correct where (d') : every bounded random variable is in $\mathbf{F}$.

50. Given any two random variables you can talk about their maximum. Given a sequence of random variables you can talk about their supremum. What if we have arbitrary collection of random variables? To understand the problem, start with an example:

    Take $(R, \mathcal{B}, P)$ where P is the standard normal probability. Fix a set $M \subset R$ which is not a Borel set. Consider the family $(I_{\{x\}} : x \in M)$. If you take the pointwise supremum then you get I_M which is not a random variable.

    Suppose that I take the family $(I_{\{x\}} : x \in R)$. If I take pointwise supremum of these random variables we get the constant function $\mathbf{1}$. But what is its worth? Is probability playing any role? Do we not feel that each of the functions we have are *essentially* just the zero random variables? Their supremum should naturally be the zero random variable. So we give up pointwise supremums. We bring in our probability to the fore.

    Now let $(\Omega, \mathcal{A}, P)$ be a probability space. Let $\mathbf{L}$ be a collection of random variables. Say that a random variable Z is *essential supremum* denoted by *esup* if the following holds: (i) $X \in \mathbf{L}$ implies $P(X \leq Z) = 1$. (Thus Z is an upper bound for all our random variables.) (ii) If W is a random variable and $X \in \mathbf{L}$ implies $P(X \leq W) = 1$ then $P(Z \leq W) = 1$ (Thus Z is the least upper bound for our random variables.)

    The plan is to show that any family $\mathbf{L}$ indeed has *esup*.

    Step 1: Argue that there is no loss in assuming $X \in \mathbf{L}$ and $Y \in \mathbf{L}$ implies $X \vee Y \in \mathbf{L}$. Now on assume this.

    Step 2: Assume that all our random variables take values in $[-1, 1]$. Put $\alpha = Sup[E(X) : X \in \mathbf{L}]$. Get X_n from $\mathbf{L}$ such that $E(X_n) \uparrow \alpha$. Put $Z = supX_n$ – here pointwise supremum. Show that this does.

    Step 3 : Consider the new class $\mathbf{L}' = \{tan^{-1}X : X \in \mathbf{L}\}$. Take the principal value and interpret $tan^{-1}(\pm\infty) = \pm\pi/2$. Apply step 2 , get Z for $\mathbf{L}'$. Show $tanZ = W$ does for $\mathbf{L}$. The only thing you need to know is that the tan^{-1} preserves order and takes you from $[-\infty, \infty]$ to $[-\pi/2, \pi/2]$; completing proof of existence of *esup*.

Step 4 : Show that *esup* of a family is unique in the sense that if Z_1 and Z_2 are two such then $P(Z_1 = Z_2) = 1$.

What if we have a σ-finite measure space? How should we define *esup*. Do they exist?

Home Work 7

51. You know that integration is finding areas – at least for nonnegative functions the integral is the area under the curve (above X-axis). This is *still* correct if properly understood.

    Let $(\Omega, \mathcal{A}, P)$ be a probability space and X be a nonnegative random variable – more specifically $0 \leq X \leq 1$. Consider the probability space $([0,1], \mathcal{B}, \lambda)$. Consider the product space $\Omega \times [0,1]$ with product σ-field and product probability $P \otimes \lambda$.

    The graph G of the random variable is the set $\{(\omega, y) : y = X(\omega)\}$. Show that G is in the product σ-field. The portion below the graph is the set $G_< = \{(\omega, y) \in \Omega \times [0,1] : y \leq X(\omega)\}$. Show that this is also in our σ-field. Show that its area, namely, $P \otimes \lambda(G_<)$ is nothing but $E(X)$.

    It is very unfortunate that we had to restrict to random variables taking values in [0,1]. We had to do that because we talked about only products of probability spaces. So let us learn about σ-finite spaces.

52. Let $(\Omega_1, \mathcal{A}_1, \mu_1)$ and $(\Omega_2, \mathcal{A}_2, \mu_2)$ be two σ-finite measure spaces. Carry out all the steps of the class and show that there is a unique σ-finite measure $\mu_1 \otimes \mu_2$ on the product space such that for every rectangle $A \times B$ its value is $\mu_1(A)\mu_2(B)$.

    Show that Fubini is still correct.

53. Let $(\Omega, \mathcal{A}, P)$ be a probability space and X be a nonnegative random variable. Considering the product space $\Omega \times [0, \infty)$ and the product measure $P \otimes \lambda$ show that $E(X)$ is nothing but the area under the graph of X. Here λ is the Lebesgue measure on $[0, \infty)$.

54. In probability we need to have not only *two* independent random variables but many more. In analysis we need not only areas but volumes too.

    For $1 \leq i \leq k$, let $(\Omega_i, \mathcal{A}_i, P_i)$ be probability spaces. Define $\Omega = \times\Omega_i$. Let $\mathcal{R}$ be the collection of measurable rectangles – that is sets of the form $\times A_i$ where for each i, A_i is in $\mathcal{A}_i$. The σ-field generated by these rectangles is denoted by $\mathcal{A} = \otimes\mathcal{A}_i$. Show that there is a unique probability $P = \otimes P_i$ on this product σ-field such that for any rectangle as above $P(\times A_i) = \Pi P_i(A_i)$. This unique probability is called the product probability – more precisely the product of the probabilities P_i. (Do it for $k = 3$ first and try to think how to do it for general k. Do not say *similarly this can be done* . You can use such phrases *only if* you have done it once and convinced that it can indeed be done similarly – not because your teacher said so!)

    State Fubini and prove.

55. Do the same thing (that is, finite products) for σ-finite measures.

56. The special case of the above exercise when each space is Real line with Borel σ-field and Lebesgue measure will provide you a σ-finite measure on R^k. This is called the Lebesgue measure on R^k. The standard densities you learnt are densities w.r.t this measure.

57. Consider a probability space $(\Omega, \mathcal{A}, P)$.

    A complex valued function Z is said to be a random variable iff the two real valued functions $Re\, Z$, and $Im\, Z$ are real valued random variables. Here $Re\, Z$ is defined as the real part of Z (at each point). Similarly $Im\, Z$ is defined. Say that Z is integrable iff both $Re\, Z$ and $Im\, Z$ are integrable. In such a case we define $E(Z) = E(Re\, Z) + E(Im\, Z)$.

    Show that the collection of complex random variables is a complex vector space and the collection of integrable complex random variables is a complex subspace. Further on this later space Expectation is a complex linear map. State and prove Lebesgue's DCT.

    Extend (integration of complex random variables) to σ-finite measure spaces.

    This exercise set is mostly generalizations – either from probability spaces to σ-finite spaces *or* from product of two spaces to finitely many *or* from real to complex random variables. This will really help you to see if you are getting the ideas or not. No need to worry because we discuss in the class in case you have any difficulty.

Home Work 8

Suppose that $(\Omega, \mathcal{A}, P)$ is a probability space.

58. Say that two subfamilies $\mathcal{C}_1$ and $\mathcal{C}_2$ contained in $\mathcal{A}$ are independent in case the following holds : $C_i \in \mathcal{C}_i$ for $i = 1, 2$ implies that $P(C_1 \cap C_2) = P(C_1)P(C_2)$.

    Suppose that the families $\mathcal{C}_i$ are fields and are independent. Show that $\sigma(\mathcal{C}_1)$ and $\sigma(\mathcal{C}_2)$ are independent.

    Do the same if I just told you that the two families are semi-fields instead of fields.

59. Suppose that X and Y are two random variables. Show that they are independent (as defined in class) if and only if the families $\sigma(X)$ and $\sigma(Y)$ are independent (as defined above).

60. Say that subfamilies $\mathcal{C}_i$, $(1 \leq i \leq n)$; contained in $\mathcal{A}$ are independent in case the following holds : $C_i \in \mathcal{C}_i$ for $1 \leq i \leq n$ implies that $P(\cap C_i) = \Pi P(C_i)$.

    Suppose that the families $\mathcal{C}_i$ are fields and are independent. Show that $\sigma(\mathcal{C}_i)$, $(1 \leq i \leq n)$; are independent.

    Do the same if I just told you that the families are semi-fields instead of fields.

61. Show that $X_1, X_2, \cdots X_n$ are independent (as defined in class) if and only if the families $\sigma(X_i)$ are independent (as defined above).

62. Suppose that the families $\mathcal{C}_i$ are σ-fields and are independent. Take disjoint subsets S and T of $[1, 2, \cdots n]$. Put $\mathcal{D}_1 = \sigma(\cup_{i\in S}\mathcal{C}_i)$ and $\mathcal{D}_2 = \sigma(\cup_{i\in T}\mathcal{C}_i)$. Show that these families are independent.

    More generally take disjoint subsets S_j contained in $[1, 2, \cdots n]$ and put $\mathcal{D}_j = \sigma(\cup_{i\in S_j}\mathcal{C}_i)$. Show that these families are independent.

63. Suppose X_1, X_2 , $\cdots$, X_n are independent random variables. Make precise and prove the following : If Y_1 is a random variable which is a function of X_i, $1 \leq i \leq n_1$ and Y_2 is a random variable which is a function of X_i , $n_1 + 1 \leq i \leq n_2$ etc, then Y_j are independent.

    For example $\frac{cosX_1.exp(X_2)}{X_3^4+1}$ and $tan^{-1}(X_4).cosX_5$ are independent. Imagine proving this using just the definition of independence via distribution functions.

64. Consider $(R, \mathcal{B}, P)$. How bad are random variables compared to continuous functions? An interesting theorem of the Russian mathematician Luzin is that every random variable is nearly a continuous function. More precisely, given a random variable and $\epsilon > 0$ we can find a continuous function f such that $P(X \neq f) < \epsilon$.

    Suppose we did it for bounded random variables. Given possibly unbounded X get M such that $P(|X| > M) < \epsilon/2$. Consider $Y = XI_{[|X|\leq M]}$. Get for Y a continuous g such that $P(Y \neq g) < \epsilon/2$. This g will do for X upto ϵ.

    Given a closed set $C \subset U$ open there is a continuous function f such that it is 1 for points in C and 0 for points outside U and $0 \leq f \leq 1$.

    Show that the result is correct for Indicator of closed set. Show that the result is true for indicator of Borel sets. Show that the result is true for simple random variables. In fact show that given simple r.v. X and $\epsilon > 0$, there is a continuous f as required and moreover f in modulus bounded by $sup|X(\omega)|$.

    Given a bounded random variable, take simple X_n uniformly converging to X. No loss in assuming that $Sup|X_n - X| < 1/2^n$. Get f_n for $X_n - X_{n-1}$ as in the above para upto $\epsilon/2^n$ and consider $\sum f_n$.

65. Show that the above result is true for Radon measures too.

66. Suppose that $X_n \to X$ *a.e.* Of course it need not converge uniformly. But how badly can it fail from uniform convergence? Not much. The convergence is nearly uniform according to the Russian Mathematician Egoroff. More precisely, given $\epsilon > 0$, we can get a set $A \in \mathcal{A}$ such that $P(A) > 1 - \epsilon$ and on the set A, X_n converges uniformly to X.

    Let $A_m = \{\omega : |X_n(\omega) - X(\omega)| < 1/2^k \text{ for all } n \geq m\}$. Get m_k with $P(A_{m_k}) > 1 - \epsilon/2^k$ and put $A = \cap_k A_{m_k}$.

Home Work 9

We have been on the highway long enough, now we are taking a turn towards probabilistic questions. It is better to look around the terrain we travelled so far, lest we take things for granted without verifying.

67. If $|X|$ is a random variable then should X be random variable?

    If the product $X.Y$ is a random variable should X and Y be random variables? Should at least one of them be r.v?

    Consider $(R^2, \mathcal{B}^2)$. Suppose that I have a set in the product space, say, A. Suppose that **all** of its sections A_{ω_1} and A^{ω_2} are available in $\mathcal{B}$. Should A be in the product σ-field $\mathcal{B}^2$?

68. Does the MCT for integrals hold if we have non-negative decreasing sequence?

    Does $EX = 0$ imply that $X = 0$ *a.e.* Or should there be at least one sample point where the random variable is zero?

    If $X_n \to X$ for every point then should we have $E(X_n) \to E(X)$.

    If $E(X_n) \to E(X)$ then should there be at least one point ω where $X_n(\omega) \to X(\omega)$?

69. Caratheodory extension can be done for not necessarily σ-finite measures. Since we have better things to do, we did not go into this. But extensions need not be unique. Consider $(R, \mathcal{B})$ and the usual field $\mathcal{F}$ consisting of finite disjoint unions of intervals of the form $(a, b]$ with $-\infty \leq a \leq b \leq +\infty$. Take μ to be zero for empty set and $+\infty$ for nonempty sets of the field. Here are two extensions: $\mu_1(B)$ is zero or infinity according as $B = \emptyset$ or not. And $\mu_2(B)$ is the number of rationals in B. One is σ-finite and the other is not. Can you think of more extensions?

70. If repeated integrals exist and equal should the double integral exist? (Consider R^2 with product Lebesgue measure and consider the random variable which is zero off A and for points in A, it is ± 1 according as the point is above the diagonal or not, where $A = (|x - y| \leq 1)$.

    It may so happen that the function is not a random variable, but repeated integrals exist. Take a set Ω with countable - cocountable σ-field and usual zero-one probability giving zero to singletons.

    Assume that the set Ω is (i) uncountable, (ii) has a linear order $\leq$ so that for each $y \in \Omega$ the initial segment $\{x : x \leq y\}$ is countable. This is not as unimaginable as it appears at first sight. The set of natural numbers N with usual order is (i) infinite, (ii) for each $y \in N$, the initial segment $\{x : x \leq y\}$ is finite.

    Consider the subset $A = \{(x, y) : x \leq y\}$ of Ω^2. Show that the sections are in the respective σ-fields. Calculate the repeated integrals.

71. Suppose that Ω is a set and $\mathcal{F}_1$ and $\mathcal{F}_2$ are two fields of subsets of Ω. Is their union (that is, the class of sets which are in one of these fields) a field? Is their intersection (that is, the class of sets which belong to both the fields) a field? What if 'field' is replaced by σ-field throughout.

Suppose that I have an infinite sequence of fields of subsets $\mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots$ of Ω. Is their union a field? What if I replace 'field' by σ-field at all/some places?

What if I pose the questions for monotone classes?

72. Suppose that I have two σ-fields of subsets of Ω, say $\mathcal{A}$ and $\mathcal{B}$. Let $\mathcal{C} = \{A \cap B : A \in \mathcal{A}; B \in \mathcal{B}\}$. Show that this class is a semifield and $\sigma(\mathcal{C}) = \mathcal{A} \vee \mathcal{B}$ where the right side means the smallest σ-field including both $\mathcal{A}$ and $\mathcal{B}$.

    Consider $(\Omega \times \Omega, \mathcal{A} \otimes \mathcal{B})$. Let D denote the diagonal and $\mathcal{D}$ denote the product σ-field restricted to D. Define a map $\pi : D \longrightarrow \Omega$ by $\pi(\omega, \omega) = \omega$. Show that this map is one to one and onto. Show that for every $S \in \mathcal{D}$, we have $\pi(S) \in \mathcal{A} \vee \mathcal{B}$ and conversely for every $S \in \mathcal{A} \vee \mathcal{B}$ we have $\pi^{-1}(S) \in \mathcal{D}$. This property is expressed by saying that $(D, \mathcal{D})$ and $(\Omega, \mathcal{A} \vee \mathcal{B})$ are isomorphic and π is an isomorphism.

73. Let $\mathcal{C}$ be a class of subsets of Ω. If $A \in \sigma(\mathcal{C})$ then show that there is a sequence of sets $(C_n : n \geq 1) \subset \mathcal{C}$ such that $A \in \sigma(C_n : n \geq 1)$.

    Suppose that $\mathcal{A}$ is a σ-field of subsets of Ω. Show that the diagonal $D \in \mathcal{A} \otimes \mathcal{A}$ iff there is a countably generated sub σ-field $\mathcal{A}_0$ of $\mathcal{A}$ such that every singleton set is in $\mathcal{A}_0$.

    More generally Suppose that $\mathcal{A}$ and $\mathcal{B}$ are σ-fields of subsets of Ω. Show that the diagonal $D \in \mathcal{A} \otimes \mathcal{B}$ iff there are countably generated sub σ-fields $\mathcal{A}_0$ of $\mathcal{A}$ and $\mathcal{B}_0$ of $\mathcal{B}$ such that every singleton set is in these sub σ-fields.

    Suppose you take the countable-cocountable σ-field on R. Show that the diagonal is not in the product σ-field.

74. Suppose that $\mathcal{C}$ is any family of subsets of Ω. Let $\mathcal{C}_1$ be the family of sets A such that $A = \emptyset$ or $A = \Omega$ or $A \in \mathcal{C}$ or $A^c \in \mathcal{C}$. Let $\mathcal{C}_2$ be the family of finite intersections of sets in $\mathcal{C}_1$. Let $\mathcal{F}$ be the family of finite disjoint unions of sets in $\mathcal{C}_2$. Show that this is the smallest field of subsets of Ω which includes the given class $\mathcal{C}$ — In other words, this is the field generated by $\mathcal{C}$.

75. Let $\mathcal{A}$ be a σ-field on Ω and X , Y be two random variables. Define the map $\varphi : \Omega \longrightarrow R^2$ by $\varphi(\omega) = (X(\omega), Y(\omega))$. Show that this is an R^2 valued random variable on Ω – that is, for every $B \in \mathcal{B}^2$, $\varphi^{-1}(B) \in \mathcal{A}$. In fact show that $\sigma(X, Y) = \{\varphi^{-1}(B) : B \in \mathcal{B}^2\}$. Generalize to finitely many random variables.

    Random variables taking values in R^n are called *random vectors*.

76. Show that $\mathcal{B}^2 = \sigma(\mathcal{O})$ where $\mathcal{O}$ is the collection of open sets in R^2. Generalize to R^n.

    The moral is, we can define the Borel σ-field on R^n either as the σ-field generated by its open sets *or* as the n-fold product of the Borel σ-field of R.

77. Recall the coin tossing space 2^ω — infinite sequences of zeros and ones with its natural σ-field and the fair coin tossing probability. Define

$$X(\epsilon) = \sum \epsilon_n / 2^n \quad \text{for } \epsilon = (\epsilon_1, \epsilon_2, \cdots) \in 2^\omega$$

Show that this is a random variable on the coin tossing space and the distribution of this random variable is uniform $[0,1]$. Show that X is onto [0,1]. If you take a number $a \in [0,1]$ how many points can be there in $X^{-1}(a)$?

Consider the coin tossing space, and define the random variables : $X_n = -1$ or $+1$ according the n-th coordinate is zero. What would they correspond to on the unit interval?

78. Suppose $(\Omega, \mathcal{A}, P)$ is a probability space. Suppose that I take a set $S \subset \Omega$ which may possibly be not in my σ-field. I realize now that the set S is important for me. I want to enlarge my σ-field so that it includes this set also. How does this new σ-field look like? Here is the answer. Show that $\sigma(\mathcal{A}, S) = \mathcal{B}$, say, is precisely

$$\mathcal{B} = \{(A_1 \cap S) \ \cup \ (A_2 \cap S^c) : A_1, A_2 \in \mathcal{A}\}$$

Suppose now you want me to extend my probability to this larger σ-field and you want the probability of S to be a number c. Can I do it and if so how? To come to grips with the problem, let us first see how large/small is S.

Put $\alpha = \sup[P(A) : A \in \mathcal{A}; A \subset S]$ and $\beta = \inf[P(A) : A \in \mathcal{A}; A \supset S]$. Clearly $\alpha \leq \beta$. Show that these are actually max and min. That is, there are indeed sets $\underline{S} \subset S \subset \overline{S}$ in $\mathcal{A}$ such that $P(\underline{S}) = \alpha$ and $P(\overline{S}) = \beta$. Show that if $c < \alpha$ or $c > \beta$ then no extension with $P(S) = c$ is possible.

Show that the sets $\underline{S}$ and $\overline{S}$ are essentially (?) unique. Now assume that $\alpha \leq c \leq \beta$. Define $\tilde{P}[(A_1 \cap S) \ \cup \ (A_2 \cap S^c)]$ where $A_1, A_2 \in \mathcal{A}$ as $c\,P(A_1 \cap \overline{S}) + (1-c)\,P(A_2 \cap (\underline{S})^c)$. Show that this is a good definition (?), it is a probability on $\mathcal{B}$, it extends P, its value for S is the given number c.

79. Show that convergence *a.e.* , convergence in probability respect addition of random variables as well as scalar multiplication. They respect multiplication of random variables too. They respect composition with continuous function, that is, if $X_n \to X$ and φ is a continuous function on R to R, then $\varphi(X_n) \to \varphi(X)$.

Show that L_p convergence respects addition and scalar multiplication. Multiplication of two L_p random variables may not be in L_p. Take Poisson probability on the real line. Exhibit, for each $p \geq 1$ a random variable which is in the L_r spaces for $r < p$ but not for $r > p$.

Home Work 10

80. Consider the coin tossing space 2^ω and for $0 \leq p \leq 1$ the probability P_p corresponding to independent tosses of a coin whose chance of heads in any toss is p. Show that these probabilities are *uniformly singular*. This means that there are disjoint sets A_p for $0 < p < 1$, such that, $P_p(A_p) = 1$ for all p.

(Try SLLN)

81. Consider a sequence of i.i.d random variables taking values ± 1 each with probability $1/2$. Let S_n be the partial sum of the first n of these variables. Show that for $t > 0$, $\quad P(|S_n| > t) \leq 2e^{-t^2/2n}$.

    (Try Markov inequality for $P(e^{S_n} > e^t)$.)

    This is called Chernoff bound – simple but very powerful. Because of the exponential term this is called exponential bound. Since this gives a bound for the probability that S_n deviates by an amount larger than t from its mean value, it is called a large deviation bound. The method of proof uses m.g.f and is hence called the mgf method.

    Use this inequality to show that $S_n/n \to 0 \quad a.e.$

    (Try Borel Cantelli carefully)

82. Consider the unit interval with Lebesgue measure (uniform distribution). Fix any integer $r \geq 2$ – usually called radix or base. Show that every number ω in the unit interval has an expansion $\sum_{n\geq 1} \omega_n r^{-n}$ where for each n, ω_n is an integer and $0 \leq \omega_i < r$. Show that the set of those numbers for which there is more than one such expansion is countable and hence has Lebesgue measure zero (How do such numbers with more than one expansion look like?) The number ω_n is called the n-th digit in the expansion of a to the base r. Let X_n be the n-th digit in the expansion to the base r. So these are random variables defined on the probability space $([0,1], \mathcal{B}, \lambda)$. Show that they are i.i.d having uniform (?) distribution on $\{0, 1, 2 \cdots, r-1\}$. Show that for almost every number in the unit interval the each of these integers appears with equal frequency in its r-expansion. Such numbers a are called r-normal numbers.

    Show that for almost every number in the unit interval the following is true : whatever be the integer $r \geq 2$; the digits $0, 1, 2, \cdots, r-1$ appear with equal frequency in its r-expansion. Such numbers are called absolutely normal numbers. (Though almost all numbers are absolutely normal, it is difficult to exhibit *one* number and show that it is absolutely normal.)

83. Let $(\Omega, \mathcal{A}, P)$ be a probability space and X_n be a sequence of independent random variables with mean zero. We showed that if $\sum Var(X_n) < \infty$ then the series $\sum X_n$ is convergent a.e. It is natural to ask if $\sum Var(X_n) = \infty$ then the series is non-convergent. The answer is : No. Try X_n with values, $+n^2$, 0, $-n^2$ with probabilities $1/n^2$, $1 - \frac{2}{n^2}$ and $1/n^2$ respectively.

    However if the random variables are uniformly bounded then it is indeed true. Take X_n independent mean zero and $\sum X_n$ converging a.e. Say for all n, $|X_n| \leq c$.

    First use Egoroff's theorem and get a set A with $P(A) > 0$ and $|S_n| \leq d$ for all n on the set A. Here S_n is the partial sum of the X sequence.

    Denote by A_n the set ($|S_m| \leq d$ for $m = 1, 2, \cdots, n$). Set $a_n = E(I_{A_n} S_n^2)$. Show that

$$a_n - a_{n-1} \geq P(A) Var(X_n) - (c+d)^2 P(A_{n-1} - A_n)$$

    Show that $\sum Var(X_n) < \infty$.

84. Suppose that X_n is a sequence of independent random variables on a space and are uniformly bounded. Show that $\sum X_n$ is convergent a.e. iff $\sum E(X_n)$ and $\sum Var(X_n)$ are convergent.

(If both the numerical series are convergent, then apply what you know to the sequence $X_n - E(X_n)$. If you know convegence a.e. of the random series then consider the product space $\Omega \times \Omega$ with product probability and apply what you know to the sequence $Z_n = U_n - V_n$ where $U_n(\omega, \omega') = X_n(\omega)$ and $V_n(\omega, \omega') = X_n(\omega')$ to get convergence of variance series. Apply what you know to $X_n - E(X_n)$ and deduce about the series $E(X_n)$.

85. The above result has an interesting special case. Suppose we consider a fixed sequence of numbers a_n. When does the series $\pm a_n$ converge for almost all choices of the signs. This is called the *random signs* problem. Here the signs are decided by independent tosses of a fair coin. This problem was first considered by the Polish mathematician Steinhaus and also by the Russian mathematician Khinchin.

To make it precise consider a sequence of i.i.d random variables Z_n each taking values ± 1 with equal probability. Show that the series $\sum a_n Z_n$ converges a.e. iff $\sum a_n^2$ is finite.

For example you know that the series $\sum 1/n$ is not convergent whereas the series $1 - 1/2 + 1/3 - 1/4 + - \cdots$ is convergent. By putting plus and minus signs you can produce convergent series as well as non-convergent series. One wonders if there is any *typical* behaviour. the above result says that for typical choice of signs the series does indeed converge.

86. You know that if $\sum P(A_n) < \infty$ then $P(limsup A_n) = 0$. It is natural to ask if $\sum P(A_n) = \infty$ then $P(limsup A_n) = 1$. Try counter examples.

However if the events are independent then this is right. That is, if A_n are independent events and $\sum P(A_n) = \infty$ then $P(limsup A_n) \neq 0$. (Apply what you know to the sequence I_{A_n}.)

Here is another way of getting at it – which avoids all the discussion about series — provided you know about infinite products. Use

$$P(\cap A_n^c) \quad = \Pi\,[1 - P(A_n)] \quad \leq e^{-\sum P(A_n)} \quad = 0$$

87. Here is the definitive answer for the convergence of infinite series of independent random variables. Suppose that X_n are independent random variables. Suppose that there is a number $c > 0$ such that denoting $A_n = (\omega : |X_n(\omega)| \leq c)$ the following three numerical series converge:

$\sum P(A_n^c)$; $\sum \int_{A_n} X_n\, dP$ and $\sum [\int_{A_n} X_n^2\, dP - (\int_{A_n} X_n\, dP)^2]$

Show that the series $\sum X_n$ converges a.e.

(Show that the sequences (X_n) and $(X_n\, I_{A_n})$ are equivalent in the sense of Khinchin and the latter sequence is bounded.)

Conversely, suppose the the series $\sum X_n$ converges a.e. Show that for *any* number $c > 0$, the above three numerical series converge.

(Try same idea as above)

This is known as the *Three series theorem* of Kolmogorov.

88. Let μ_1 and μ_2 be a probabilities on R. Define for every Borel set B, $\mu_1 * \mu_2(B) = \int \mu_1(B - x)\, d\mu_2(x)$. Show that this is again a probability on R. This is called convolution of μ_1 and μ_2. Show that if X and Y are independent random variables with distributions μ_1 and μ_2 , then $X + Y$ has the distribution $\mu_1 * \mu_2$. Show that this binary operation on the collection of probabilities on R is commutative and associative. Show that c.f of the convolution is the product of c.f.s.

If μ is a probability then $\tilde{\mu}(B) = \mu(-B)$ is again a probability. If $X \sim \mu$ then $-X \sim \tilde{\mu}$. Recall that μ is said to be a continuous (or non-atomic) probability if every singleton has mass zero.

Show that μ is continuous iff $\mu * \tilde{\mu}$ has no mass at zero.

Show that the following are equivalent :

(i) μ is continuous.

(ii) $\mu * \tilde{\mu}$ is continuous.

(iii) $\mu * \tilde{\mu}$ has no mass at zero.

(iv) $Lim_{T\to\infty} \frac{1}{T} \int_0^T |\varphi(t)|^2\, dt = 0$.

(v) $Lim_{T\to\infty} \frac{1}{T} \int_0^T |\varphi(t)|\, dt = 0$.

Criterion (v) is called Wiener's criterion for continuity of the distribution function.

0. Getting Started : Thm 1: Impossibility of Model; Thm 2 : Possibiliy of model

1. Probability Spaces : Thm 3 : getting σ-fields; Thm 4: Monotone class theorem; Thm 5: Continuity of probability; Thm 6: A uniqueness theorem for probabilities; Thm 7: Caratheodory extension theorem; Thm 8: Basic existence theorem of probability;

2. Random Variables : Thm 9 : Getting σ-fields; Thm 10 : Structure of random variables;

3. Expectation : Thm 11 : Properties of expectations;

4. Fine Tuning : Infinities; σ-finite measures; complex random variables; almost everywhere notion;

5. Product Spaces : Thm 12 : Product probability; Thm 13 : Fubini;

6.Inequalities : Thm 14 : Basic inequalities;

7. Convergence : Thm 15 : relations between convergence notions; Thm 16 : Series of independent random variables; Thm 17 : SLLN; Thm 18 : Kolmogorov's $0-1$ law;

8. Characterstic Functions : Thm 19 : Basic results; Thm 20 : CLT;

9. Extras : Thm 21: Radon-Nikodym and Lebesgue decomposition; Thm22 : Jessen-Wintner;

(HW 1: number of sets, Cantor set and df; Gdelta etc; *2: df and prob, regularity, 0-1 prob, cg, plane; *3 Lebesgue on R , df and rdf ;*4 random var, infinities, countable sigma fields *5 Expectation BStat def coin tossing space *6 Radon Nikodym, Change of variable, completion, C infinity functions, ess sup; *7 Integral as area, higher dim Fubini, complex functions *8 Independence, Luzin, Egoroff; *9 Counter ex, measure on coin tossing space, Extending for one more set Exactly in L_r; *10 Chernoff, absolutely normal numbers, three series, random signs, Converse of Borel Cantelli, convolution, cf.

Must put: Lp completeness and Lp separability/ Jordan Hahn/ Diagonal in product/ two notions of indep/Kol consistency and counter ex/Pacific Jl mustar/Analytic sets in completion/set Omega and generation of Borel sets/ Levy discrete?/Cramer Wold?/

23 Jordan Hahn 24 K's consistency 25 Ω and no nonatomic probability.